LECTURE NOTES ON ERGODIC THEORY

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1. The setup of Ergodic Theory

Ergodic theory can be roughly described as the study of **measure preserving dynamical systems.**

More precisely, by a **(discrete time) dynamical system** we mean a map $T\colon X \to X$, where X is its **phase space** and the dynamics is given by the iteration of T: $T^0 = id_X$ and $T^{n+1} := T \circ T^n$, for every $n \geqslant 0$. When T is bijective, one can consider the "past trajectories" taking negative iterates that are defined by $T^{-n} := (T^{-1})^n$, for every $n \geqslant 0$.

We say that $T\colon X \circlearrowleft$ is a **measurable** system when X is a endowed with a σ-algebra $\mathscr{B} \subset 2^X$ and T is a $\mathscr{B}$-measurable map.

When μ is a measure on the measurable space $(X, \mathscr{B})$ and $T\colon X \circlearrowleft$ is a measurable map, we say that T **preserves** μ, or that μ is a **T-invariant measure**, when it holds

$$\mu(A) = \mu\big(T^{-1}(A)\big), \quad \forall A \in \mathscr{B}.$$

So, a **measure preserving system** is the quadruple $T\colon (X, \mathscr{B}, \mu) \circlearrowleft$ where $T\colon X \circlearrowleft$ is a $\mathscr{B}$-measurable map and μ is a T-invariant **probability** measure, i.e. where $\mu(X) = 1$.

Analogous concepts can be defined for continuous time dynamical systems, also called **semi-flows** and **flows**, and more generally, for (measurable) actions of semi-groups and groups.

Our main references for the course are the books of Viana and Oliveira [VO16] and Mañé [Mañ87].

2. Motivations

2.1. **Dynamical systems with a natural invariant measure.** There are many classical dynamical systems, which are used to model natural phenomena, that already arise with a "natural" invariant (finite) measure. From the historical point of view, one of the most important examples are Hamiltonian systems that are used to described the evolution of conservative systems in Classical Mechanics.

So, the first main question in ergodic theory could be briefly and roughly summarized as follows:

> *What dynamical information can be extracted from the fact that a dynamical system leaves invariant a certain measure?*

2.2. **Statistical study of dynamical systems.** A second motivation to study measure preserving systems comes from the analysis of *statistical aspects* of dynamical systems. In its simplest form, let us suppose a dynamical system $f\colon X \circlearrowleft$ is given and for certain subset $A \subset X$ we want to analyze the asymptotic to behavior (i.e. as $n \to \infty$) of the *mean visiting time function to A* $\tau_A^{(n)}\colon X \to [0,1]$ given by

$$\tau_A^{(n)}(x) := \frac{\sharp\left\{0 \leqslant i \leqslant n-1 : f^i(x) \in A\right\}}{n}, \quad \forall x \in X,\ \forall n \in \mathbb{N}.$$

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As we shall see along the course, the asymptotic behavior of functions $\left(\tau_A^{(n)}\right)_{n\geqslant 1}$ for most subsets A is determined the whole set of f-invariant measures.

2.3. **Applications to combinatorics and number theory.** As third motivation, we will see that ideas and techniques of ergodic theory can be applied to get some nice results on combinatorics and number theory. The main reference for this part of the course in the book of Einsiedler and Ward [EW11].

3. Measure theory

3.1. **Measurable spaces.** All along these notes, X will denote a non-empty set. We shall write 2^X for the set of subsets of X. A **σ-algebra** on X is a family $\mathcal{B} \subset 2^X$ such hat $\varnothing, X \in \mathcal{B}$ and given $A_1, A_2, \ldots \in \mathcal{B}$, it holds $\bigcup_{n\geqslant 1} A_n \in \mathcal{B}$. A **measurable space** is a pair $(X, \mathcal{B})$ where X is a non-empty set and $\mathcal{B}$ is a σ-algebra on X.

There is a natural partial order among σ-algebras of X given by set inclusion. Invoking this order, given any family $\mathcal{C} \subset 2^X$ one can define the **σ-algebra generated by $\mathcal{C}$**, which will be denoted by $\sigma(\mathcal{C})$, as the smallest σ-algebra that contains $\mathcal{C}$.

When X is a topological space and $\tau \subset 2^X$ is the topology of X, i.e. τ is the family of open subsets in X, the σ-algebra $\sigma(\tau)$ generated by τ is called the **Borel σ-algebra** of X is usually denoted by $\mathcal{B}(X)$.

Given a measurable space $(X, \mathcal{B})$, and an arbitrary map $f\colon X \to Y$, one can define the **push-forward σ-algebra** $f_\star\mathcal{B}$ on Y by

$$f_\star\mathcal{B} := \left\{E \subset Y : f^{-1}(E) \in \mathcal{B}\right\}. \tag{1}$$

On the other hand, given another map $h\colon Z \to X$, we defined the **pull-back σ-algebra** $f^\star\mathcal{B}$ on Z by

$$f^\star\mathcal{B} := \left\{E \subset Z : \exists A \in \mathscr{B},\ f^{-1}(A) = E\right\}. \tag{2}$$

Given two measurable spaces $(X, \mathcal{B})$ and $(Y, \mathcal{C})$, a map $f\colon X \to Y$ is said to be **measurable** when $\mathcal{C} \subset f_\star\mathcal{B}$.

Exercise 1. *If $(X, \mathcal{B})$ and $(Y, \mathcal{C})$ are measurable spaces and $f\colon X \to Y$, show that $f^\star\mathcal{C} \subset \mathcal{B}$ if and only if $\mathcal{C} \subset f_\star\mathcal{B}$.*

A **measure** on a measurable space $(X, \mathcal{B})$ is map $\mu\colon \mathcal{B} \to [0, +\infty]$ such that $\mu(\varnothing) = 0$, $\mu(X) > 0$ and

$$\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n),$$

for every sequence of pairwise disjoint sets $A_1, A_2, \ldots \in \mathcal{B}$. The measure μ on $(X, \mathcal{B})$ is said to be **finite** when $\mu(X) < \infty$; a **probability** measure when $\mu(X) = 1$; and **σ-finite** when there exists a sequence $X_1, X_2, \ldots \in \mathcal{B}$ such that $\mu(X_n) < \infty$ for every $n \geqslant 1$ and $X = \bigcup_{n\geqslant 1} X_n$.

Given a measure space $(X, \mathcal{B}, \mu)$, we say that two sets $A, B \in \mathcal{B}$ are **equal modulo μ** when $\mu(A \triangle B) = 0$, where $A \triangle B := (A \backslash B) \cup (B \backslash A)$.

Exercise 2. *Let $(X, \mathcal{B}, \mu)$ be a finite measure space. Show that the relation $\sim$ on $\mathcal{B}$ given by $A \sim B$ if and only if A and B are equal modulo μ is an equivalence relation. Then, let us consider the function $d_\mu\colon \mathcal{B}/\!\sim \times \mathcal{B}/\!\sim \to \mathbb{R}$ given by*

$$d_\mu([A], [B]) := \mu(A \triangle B), \quad \forall A, B \in \mathscr{B},$$

and where $[A], [B]$ denotes the equivalence classes of A and B, respectively. Show that the function d_μ is well defined and is a distance on $\mathcal{B}/\!\sim$.

A **measure space** is triple $(X, \mathcal{B}, \mu)$, where $(X, \mathcal{B})$ is a measurable space and μ is a measure on it. The σ-algebra $\mathcal{B}_0^\mu \subset \mathcal{B}$ of **μ-null sets** is given by

$$\mathcal{B}_0^\mu := \{N \in \mathcal{B} : \mu(N) = 0\}.$$

We say that the measure space $(X, \mathcal{B}, \mu)$ is **complete** when for every $N \in \mathcal{B}_0^\mu$ and any $N' \subset N$, it holds $N \in \mathcal{B}$.

Every measure space $(X, \mathcal{B}, \mu)$ admits a unique **completion:** it is a complete metric space $(X, \overline{\mathcal{B}}, \bar{\mu})$ such that

$$\overline{\mathcal{B}} := \left\{ B \cup N_1 \backslash N_2 : B \in \mathcal{B},\ \exists N \in \mathcal{B}_0^\mu,\ N_1, N_2 \subset N \right\},$$

and

$$\bar{\mu}(B \cup N_1 \backslash N_2) := \mu(B), \quad \forall B \in \mathcal{B},\ \forall N \in \mathcal{B}_0^\mu,\ \forall N_1, N_2 \subset N.$$

So, in order to avoid unnecessary technical difficulties, from now on we will assume all our measure spaces are complete.

Given a measure space $(X, \mathcal{B}, \mu)$ and a map $f\colon X \to Y$, we can define the **push-forward measure** $f_\star\mu$ on the measurable space $(Y, f_\star\mathcal{B})$ by

$$f_\star\mu(A) := \mu\big(f^{-1}(A)\big), \quad \forall A \in f_\star\mathcal{B}. \tag{3}$$

Exercise 3. *Let $(X, \mathcal{B}, \mu)$ be a measure space, $(Y, \mathcal{C})$ a measurable space and $f\colon X \to Y$ a measurable map. The it holds*

$$\int_X \phi \circ f \, \mathrm{d}\mu = \int_Y \phi \, \mathrm{d} f_\star\mu,$$

for any measurbale function $\phi\colon Y \to [0, \infty]$.

It is important to remark that, in general, it is not possible to pulled measures back. In fact, it is only possible to do it when the corresponding map is invertible.

Given two measure spaces $(X, \mathcal{B}, \mu)$ and $(Y, \mathcal{C}, \nu)$, a **measure preserving map** or a **metric homomorphism** is a measurable map $h\colon X \to Y$ such that $h_\star\mu = \nu$. Observe that if we were pedantically formal, taking into account that the domain of $h_\star\mu$ is $h_\star\mathcal{B} \supset \mathcal{C}$, last equality should be written as $h_\star\mu\big|_{\mathcal{C}} = \nu$. In particular, a map $f\colon X \to X$ is said to be a **metric endomorphism** or a measure preserving map when f is measurable and $f_\star\mu = \mu$. In such a case, we say that μ is an **f-invariant measure.**

A measure preserving map $h\colon (X, \mathcal{B}, \mu) \to (Y, \mathcal{C}, \nu)$ is said to be a **measure isomorphism** when there exist measurable sets $X' \in \mathcal{B}$ and $Y' \in \mathcal{C}$ such that $\mu(X \backslash X') = \nu(Y \backslash Y') = 0$, $h\big|_{X'}\colon X' \to Y'$ is a bijection and $h^{-1}\big|_{Y'}$ is a measurable map.

Finally, we will say that a measure preserving map $f\colon (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ is an **automorphism** or an **invertible measure preserving map** when f is a metric isomorphism.

3.2. **Borel and Lebesgue spaces.** Let M denote a separable metric space and let $\mathscr{B}_M$ be its Borel σ-algebra, i.e. $\mathscr{B}_M$ is the smallest σ-algebra containing every open subset.

A probability space $(X, \mathscr{B}, \mu)$ is said to be a **Borel space** when there is a Borel probability measure μ_M on $(M, \mathscr{B}_M)$ such that $(X, \mathscr{B}, \mu)$ is isomorphic to $(M, \mathscr{B}_M, \mu_M)$.

Observe that if $(X, \mathscr{B}, \mu)$ is Borel space, then there exists a countable family $\mathcal{A} \subset \mathscr{B}$ such that $\sigma(\mathcal{A}) = \mathscr{B}$, i.e. the σ-algebra $\mathscr{B}$ is countably generated.

Finally, we say that the probability space $(X, \mathscr{B}, \mu)$ is a **Lebesgue space** when it is complete and is isomorphic to the completion of a Borel probability space. See [EW11, Appendix A. 6] for more details about Borel and Lebesgue spaces.

3.3. **Smooth measures.** Let M be a finite-dimensional compact oriented smooth d-manifold M and ω be a smooth **volume form**, i.e. ω is a smooth d-form such that for every $x \in M$ and every positively oriented base $\{X_1, X_2, \dots, X_d\}$ of the tangent space T_xM, it holds $\omega(X_1, \dots, X_d) > 0$.

Then, we can define the continuous linear functional $\Lambda_\omega \colon C^\infty(M, \mathbb{R}) \to \mathbb{R}$ by

$$\Lambda_\omega(\phi) := \int \phi\omega, \quad \forall \phi \in C^\infty(M, \mathbb{R}).$$

One can easily check that Λ_ω can be uniquely extended to a continuous linear functional $\overline{\Lambda_\omega} \colon C^0(M, \mathbb{R}) \to \mathbb{R}$, and by Riesz' representation theorem we conclude that there exists a finite Borel measure μ_ω such that

$$\overline{\Lambda_\omega}(\phi) = \int_M \phi \, \mathrm{d}\mu_\omega, \quad \forall \phi \in C^0(M, \mathbb{R}).$$

The measure μ_ω is said to be a **smooth measure.**

Exercise 4. *Can we define smooth measures on non-compact oriented manifolds? Can smooth measures be finite on non-compact manifolds? Smooth measures can be defined on non-orientable manifolds?*

4. Recurrence on finite measure spaces

In this section we shall analyze the fact that finiteness of invariant measures imposes recurrence of the dynamical system.

Our first result is the celebrated

Theorem 4.1 (Poincaré recurrence theorem)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space and $f \colon (X, \mathcal{B}, \mu) \circlearrowleft$ be a measure preserving map. Let $A \in \mathscr{B}$ be a measurable set such that $\mu(A) > 0$. Then, there exists $n \geqslant 1$ such that*

$$\mu\big(A \cap f^{-n}(A)\big) > 0.$$

Proof. Let us suppose that $A, f^{-1}(A), f^{-2}(A), \dots$ are essentially disjoint, i.e. suppose that

$$\mu\big(f^{-i}(A) \cap f^{-j}(A)\big) = 0, \quad \forall i, j \in \mathbb{N}_0, \ i \neq j. \tag{4}$$

Let us define the sets $A_0 := A$, and $A_n := f^{-n}(A) \backslash \Big(\bigcup_{i=0}^{n-1} f^{-i}(A) \Big)$, for every $n \geqslant 1$. Then, the sets $A_0, A_1, A_2, \dots$ are pairwise disjoint, and by (4), we have

$$\mu\big(f^{-n}(A)\big) \geqslant \mu(A_n) \geqslant \mu(f^{-n}(A)) - \sum_{i=0}^{n-1} \mu\big(f^{-n}(A) \cap f^{-i}(A)\big) = \mu\big(f^{-n}(A)\big),$$

for every $n \geqslant 1$, and hence it holds $\mu\big(f^{-n}(A)\big) = \mu(A_n)$, for any $n \geqslant 0$. Since μ is f-invariant, this implies $\mu(A_n) = \mu(A) > 0$, for every $n \geqslant 0$, getting a contradiction because

$$1 = \mu(X) \geqslant \mu\Big(\bigsqcup_{i=0}^{\infty} A_n \Big) = \sum_{i=0}^{\infty} \mu(A_n) = \sum_{i=0}^{\infty} \mu(A) = \infty.$$

So, assumption (4) does not hold and hence, there exist $i, j \in \mathbb{N}$, with $i < j$ such that

$$0 < \mu\big(f^{-i}(A) \cap f^{-j}(A)\big) = \mu\Big(f^{-i}\big(A \cap f^{i-j}(A)\big)\Big) = \mu(A \cap f^{-(j-i)}(A)).$$

□

There are a lot of important results that can be proven as a consequence of Theorem 4.1. The firs one is indeed a very simple improvement of Theorem 4.1:

Corollary 4.2. *Let $f\colon (X,\mathcal{B},\mu) \circlearrowleft$ and $A \in \mathcal{B}$ be as in Theorem 4.1. Then, there is a natural number $n \leqslant \frac{1}{\mu(A)}$ such that*

$$\mu\big(A \cap f^{-n}(A)\big) > 0.$$

Exercise 5. *Prove Corollary 4.2.*

Corollary 4.3. *Let $f\colon (X,\mathcal{B},\mu) \circlearrowleft$ and $A \in \mathcal{B}$ as in Theorem 4.1. Then the set*

$$A_f := \{x \in A : \exists n \geqslant 1,\ f^n(x) \in A\}$$

belongs to $\mathcal{B}$ and $\mu(A \backslash A_f) = 0$.

Proof. First notice that

$$A_f = \bigcup_{n\geqslant 1} \big(A \cap f^{-n}(A)\big).$$

So, $A_f \in \mathcal{B}$.

Then, let us define $A' := A\backslash A_f$. Observe that $A' \cap f^{-n}(A') = \varnothing$, for every $n \geqslant 1$. Hence, we have that $f^{-m}(A') \cap f^{-n}(A') = \varnothing$, whenever $0 \leqslant m < n$. By Theorem 4.1, we conclude that $\mu(A') = 0$. □

Then notice that Corollary 4.3 can be considerably improved:

Corollary 4.4. *Let $f\colon (X,\mathcal{B},\mu) \circlearrowleft$ and $A \in \mathcal{B}$ as in Theorem 4.1. Then the set*

$$A_\infty := \big\{x \in A : \exists n_j \uparrow \infty,\ f^{n_j}(x) \in A, \forall j \geqslant 1\big\}$$

belongs to $\mathcal{B}$ and $\mu(A\backslash A_\infty) = 0$.

Exercise 6. *Prove Corollary 4.4.*

4.1. **Topological recurrence.** Let X be a topological space and $f\colon X \circlearrowleft$ be an arbitrary map. We say that a point $x_0 \in X$ is **recurrent** for f when given any neighborhood V of x_0, there is a natural number $n \geqslant 1$ such that $f^n(x_0) \in V$.

Recall that a **base** for the topology of X is a family of open subsets $\mathcal{V}$ such that every open subset of X can be writte as the union of elements of $\mathcal{V}$.

Then we have the following topological version of Poincaré recurrence theorem:

Theorem 4.5. *Let X be a topological space admitting a countable base for its topology and let $\mathcal{B}(X)$ denote its Borel σ-algebra. Let μ be a Borel probability measure on X (i.e. a probability measure on $(X,\mathcal{B}(X))$) and $f\colon (X,\mathcal{B}(X),\mu) \circlearrowleft$ a measure preserving transformation. Then, μ-almost every point is recurrent for f.*

Proof. Let $\mathcal{U} := \{U_n : n \in \mathbb{N}\}$ be a base for the topology of X. For each $n \in \mathbb{N}$ let us consider the set $W_n := \{x \in U_n : f^k(x) \notin U_n,\ \forall k \geqslant 1\}$. By Corollary 4.3 we know that $\mu(W_n) = 0$, for every $n \geqslant 1$. Hence, if we write $W := \bigcup_{n\geqslant 1} W_n$, we have $\mu(W) = 0$. Then we claim that every point in $X\backslash W$ is recurrent for f. In fact, let x an arbitrary point of $X\backslash W$ and V any neighborhood of x. Since $\mathcal{U}$ is a base for the topology of X, there is a natural number n such that $x \in U_n \subset V$. Since $x \notin W$, we have $x \notin W_n$ and then, there is $k \geqslant 1$ such that $f^k(x) \in U_n \subset V$. Since V was an arbitrary neighborhood, we conclude x is recurrent for f, as desired. □

5. Some examples of measure preserving transformations

5.1. **Periodic orbits.** Let X be an arbitrary set and $z \in X$ an arbitrary point. Then we can define the δ_z on the measurable space $(X, 2^X)$ by

$$\delta_z(A) := \begin{cases} 1, & \text{if } z \in A; \\ 0, & \text{otherwise.} \end{cases}$$

Now, if $f\colon X \circlearrowleft$ is any map and there exists $p \in X$ which is **periodic** for f, i.e. there exists $k \in \mathbb{N}$ such that $f^k(z) = z$, then the **periodic measure**

$$\nu_p := \frac{1}{k}\sum_{j=0}^{k-1} \delta_{f^j(p)} = \frac{1}{k}\sum_{j=0}^{k-1} f^j_\star \delta_p$$

is f-invariant.

5.1.1. *Homeomorphisms of the interval.* Using periodic measures and Poincare's recurrence theorem (Theorem 4.5) one can easily determine all invariant measures for homeomorphisms of a compact interval... at least for homeomorphisms with finitely many periodic orbits.

In fact, if $I \subset \mathbb{R}$ denotes a compact interval and $f\colon I \circlearrowleft$ is an orientation-preserving homeomorphism (i.e. f is an strictly increasing map), then one can easily show that $x \in I$ is a recurrent point for f if and only if $f(x) = x$, and moreover, the set of fixed points $\mathrm{Fix}(f) := \{w \in I : f(w) = w\}$ is non-empty.

On the other hand, if f is orientation-reversing, i.e. f is strictly decreasing, the $\mathrm{Fix}(f)$ has exactly one point and any other recurrent point is periodic of period 2.

In any case, if f has finitely many periodic orbits, by Theorem 4.5 one have that any f-invariant Borel probability measure is indeed a convex combination of periodic ones. What can we say about the characterization of f-invariant measures when f has infinitely many periodic orbits?

5.2. **Expanding maps of the interval.** Let $k \in \mathbb{N}$ be an arbitrary number and consider the map $f\colon [0,1] \circlearrowleft$ given by

$$f(x) := kx - \lfloor kx \rfloor, \quad \forall x \in [0,1]$$

where $\lfloor x \rfloor := \max\{\ell \in \mathbb{Z} : \ell \leqslant x\}$ denotes the *integer part* of x.

We claim that the Lebesgue measure m on $[0,1]$ is f-invariant. If $J \subset [0,1]$ is a subinterval, then one can easily check that $f_k^{-1}(J)$ is the disjoint union of k intervals $J_1, \dots, J_k \subset [0,1]$ with $m(J_i) = \frac{1}{k} m(J)$, for each $i \in \{1, \dots, k\}$. So, we conclude

$$m\big(f^{-1}(J)\big) = m\Big(\bigsqcup_{i=1}^{k} J_i\Big) = \sum_{i=1}^{k} m(J_i) = \sum_{i=1}^{k} \frac{1}{k} m(J) = m(J).$$

Thus, we have proved that the two probability measures m and $f_\star m$ coincide on the subintervals of $[0,1]$. In order to conclude that they are indeed the same, we invoke the following general result:

Theorem 5.1 (Uniqueness of measures, Theorem 5.7 in [Sch17])**.** *Let μ and ν two measures on the measurable space $(X, \mathcal{B})$. Let $\mathscr{C} \subset 2^X$ be a family that generates $\mathcal{B}$, i.e. $\mathcal{B} = \sigma(\mathscr{C})$, and suppose that*

- *if $A, B \in \mathcal{C} \implies A \cap B \in \mathscr{C}$;*
- *there exists $(C_n)_{n\geqslant 1} \subset \mathscr{C}$ such that $C_n \uparrow X$ (i.e. $\bigcup_{n\geqslant 1} C_n = X$) with $\mu(C_n) = \nu(C_n) < \infty$, for every $n \geqslant 1$.*

Then, $\mu = \nu$.

5.3. **Circle rotations.** Given any $\alpha \in \mathbb{R}$, one can define the map $T_\alpha\colon [0,1] \circlearrowleft$ by

$$T_\alpha(x) := x + \alpha - \lfloor x + \alpha \rfloor, \quad \forall x \in [0,1].$$

We claim that the Lebesgue measure m on $[0,1]$ is again T_α-invariant, for every $\alpha \in \mathbb{R}$. In fact, it is enough to consider the case where $\alpha \in (0,1)$ and then, one can

easily show that given $[a,b] \subset [0,1]$, it holds

$$T_\alpha^{-1}[a,b] = [a-\alpha, b-\alpha], \quad \text{when } \alpha \leqslant a \leqslant b \leqslant 1;$$
$$T_\alpha^{-1}[a,b] = [a-\alpha+1, b-\alpha+1], \quad \text{when } 0 \leqslant a \leqslant b < \alpha;$$
$$T_\alpha^{-1}[a,b] = [a-\alpha+1, 1) \cap [0, b-\alpha], \quad \text{when } 0 \leqslant a < \alpha \leqslant b \leqslant 1.$$

Circle rotations are particular examples of translations on compact groups.

5.4. **Translations on locally compact groups.** A **topological group** is a ternary $(G, \mathscr{T}, \cdot)$ where G is a non-empty set, $\mathscr{T}$ is a Hausdorff topology on G and $(G,\cdot)$ is group such that the map $G \times G \ni (g,h) \mapsto g \cdot h^{-1} \in G$ is a continuous map.

Analogously, a **Lie group** is a ternary $(G, \mathscr{C}, \cdot)$ where G is a non-empty set, $\mathscr{C}$ is a finite dimensional C^∞ Hausdorff structure on G, $(G,\cdot)$ is a group and the map $G \times G \ni (g,h) \mapsto g \cdot h^{-1} \in G$ is C^∞. Observe that any Lie group is a locally compact topological group.

Given topological group G and any element $g \in G$, one can define the **left and right translations** by $L_g \colon G \ni h \mapsto gh \in G$ and $R_g \colon G \ni h \mapsto hg^{-1} \in G$, respectively. Notice that L_g and R_g are homeomorphism on G; and when G is a Lie group, they are C^∞ diffeomorphisms.

Definition 1 (Haar measure)**.** Given a locally compact group G, a **(left) Haar measure** is a Radon measure μ on G (i.e. μ is Borel and $\mu(K) < \infty$, for every compact $K \subset G$) such that μ is L_g-invariant, for every $g \in G$, and there is a Borel set $E \subset G$ such that $\mu(E) > 0$.

Theorem 5.2 (Existence and characterization of Haar measures)**.** *If G is locally compact group, then there exists a left Haar measure μ_G. The support of μ_G is G (i.e. $\mu_G(U) > 0$, for every non-empty open subset $U \subset G$) and $\mu_G(G) < \infty$ if and only if G is compact.*

On the other hand, if ν is another left Haar measure, then there exists $c > 0$ such that

$$\mu(E) = c\nu(E), \quad \forall E \in \mathcal{B}(G).$$

Proof. See for instance Halmos [Hal50, §58] □

Exercise 7. *If G is Lie group, then show that any left Haar measure is a smooth measures (as defined in §3.3).*

Example 5.3 (Tori)**.** The space $\mathbb{R}^d$ is an abelian Lie group and $\mathbb{Z}^d < \mathbb{R}^d$ is a (normal) subgroup. Hence, $\mathbb{T}^d := \mathbb{R}^d/\mathbb{Z}^d$ is a compact abelian Lie group, and there exists a unique left Haar measure λ^d which is a probability measure. The measure λ^d is left and right invariant is usually called the Haar or the Lebesgue measure on $\mathbb{T}^d$.

Given a topological group G, we say that a map $f \colon G \circlearrowleft$ is a **topological group endomorphism** (or just an **endomorphism** for short) when f is continuous and $f(xy) = f(x)f(y)$, for every $x, y \in G$. An endomorphism $f \colon G \circlearrowleft$ is said to be a **topological group automorphism** (or just an **automorphism**) when f is bijective and the map f^{-1} is an endomorphism as well.

Exercise 8. *Let G be a compact topological group, μ be a the left Haar probability measure and $f \colon G \circlearrowleft$ be an automorphism.*

(a) Show that μ is a right Haar measure too, i.e. it is invariant by every right translation.
(b) Show that μ is an f-invariant measure.
(c) What happend when f is just an endomorphism?
(d) And what happend when G is non-compact and just locally compact?

5.5. **The Gauss map.** Let us consider the map $G\colon (0,1) \to [0,1)$ given by

$$G(x) := \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor, \quad \forall x \in (0,1).$$

Exercise 9. *Show that given $x \in (0,1)$ there exists $n \in \mathbb{N}$ such that $G^i(x) \neq 0$, for every $i \in \{0,1,\dots,n-1\}$ and $G^n(x) = 0$ if and only if $x \in \mathbb{Q}$.*

So, if we define $X = (0,1)\backslash\mathbb{Q}$, then it holds $G\big|_X\colon X \circlearrowleft$.

A remarkable fact is that there exists a smooth probability measure μ on $(0,1)$ (or on X) which is G-invariant. In fact we have

Theorem 5.4. *The measure μ given by*

$$\mu(E) := \frac{1}{\log 2} \int_E \frac{\mathrm{d}x}{1+x}, \quad \forall E \in \mathcal{B}(0,1),$$

is a G-invariant Borel probability measure.

Proof. It clearly holds $\mu(0,1) = 1$. Let us prove that μ is G-invariant. For each $n \in \mathbb{N}$, consider the interval

$$I_n := \left(\frac{1}{n+1}, \frac{1}{n}\right), \quad \forall n \in \mathbb{N}.$$

Observe that $G_n := G\big|_{I_n}\colon I_n \to (0,1)$ is a C^∞ diffeomorphism, for each $n \in \mathbb{N}$, and $m\Big(\bigsqcup_n I_n\Big) = m(0,1) = 1$, where m denotes the Lebesgue measure on $(0,1)$. Given any $\rho \in L^1((0,1),m)$, with $\rho \geqslant 0$ m-almost everywhere, let us consider the measure μ_ρ given by

$$\mu_\rho(E) := \int_E \rho \,\mathrm{d}m, \quad \forall E \in \mathscr{B}(0,1).$$

If we define the so called *Perron-Frobenious operator* $\mathcal{L}$ by

$$\mathcal{L}\rho(y) := \sum_{k=1}^{\infty} \frac{\rho(G_k^{-1}(y))}{G_k'\big(G_k^{-1}(y)\big)}, \quad \forall y \in \bigsqcup_{k\geqslant 1} I_k,$$

we claim that

$$G_\star\mu_\rho(E) = \mu_{\mathcal{L}\rho}(E) = \int_E \mathcal{L}\rho \,\mathrm{d}m, \quad \forall E \in \mathscr{B}(0,1).$$

In order to prove this, let E be an arbitrary Borel subset of $(0,1)$ and let us write $\mathbb{1}_E$ to denote its indicator function. By the classical change of variable formula we have

$$\begin{aligned}
\mu_\rho\big(G^{-1}(E)\big) &= \int_0^1 \mathbb{1}_{G^{-1}(E)}(x)\rho(x)\,\mathrm{d}m(x) = \int_0^1 \mathbb{1}_E\big(G(x)\big)\rho(x)\,\mathrm{d}m(x) \\
&= \sum_{k=1}^{\infty} \int_{I_n} \mathbb{1}_E\big(G(x)\big)\rho(x)\,\mathrm{d}m(x) \\
&= \sum_{k=1}^{\infty} \int_0^1 \mathbb{1}_E(y)\rho\big(G_k^{-1}(y)\big)\big(G_k^{-1}\big)'(y)\,\mathrm{d}m(y) \\
&= \sum_{k=1}^{\infty} \int_0^1 \mathbb{1}_E(y)\frac{\rho\big(G_k^{-1}(y)\big)}{G_k'\big(G_k^{-1}(y)\big)}\,\mathrm{d}m(y) \\
&= \int_0^1 \mathbb{1}_E(y)\mathcal{L}(\rho)(y)\,\mathrm{d}m(y) = \mu_{\mathcal{L}\rho}(E),
\end{aligned}$$

where the last line of the chain of equalities follows by classical Levi's monotone convergence theorem.

So, if we consider $\rho(x) := \frac{1}{1+x}$, the fact that μ_ρ is a G-invariant (finite) Borel measure follows from the fact that $\mathcal{L}\rho = \rho$ and this easily follows from a straightforward computation. □

5.6. **Invariant smooth measures.** Let M and N be a smooth manifolds, α be a k-form on N and $f\colon M \to N$ be a C^1 map. The **pullback** of α by f is the k-form $f^\star\alpha$ on M given by

$$f^\star\alpha(X_1, X_2, \dots, X_k) := \alpha\big(Df_x(X_1), Df_x(X_2), \dots, Df_x(X_k)\big),$$

for all $x \in M$, and every $X_1, \dots, X_k \in T_xM$.

Exercise 10. *Let M be a compact (oriented) smooth manifold, ω a smooth volume form and $f\colon M \circlearrowleft$ be a C^1 diffeomorphism. Then, show that the induced smooth measure μ_ω is f-invariant if and only if $f^\star\omega = \omega$.*

On the other hand, let X be smooth vector field on the compact manifold M. Let us write $\Phi_X\colon \mathbb{R} \times M \to M$ for the flow induced by X. We say that a Borel measure μ on M is X-invariant (or Φ_X-invariant) when it is Φ_X^t-invariant, for every $t \in \mathbb{R}$, where $\Phi_X^t(z) = \Phi_X(z, t)$, for every $z \in M$ and any $t \in \mathbb{R}$.

Then, given a smooth k-form α on M, one defines the *Lie derivative* of α with respect to X by

$$\mathcal{L}_X\alpha := \lim_{t\to 0} \frac{\big(\Phi_X^t\big)^\star\alpha - \alpha}{t}.$$

Exercise 11. *Show that given a smooth vector field X on the compact oriented manifold M, the induced smooth measure μ_ω is X-invariant if and only if $\mathcal{L}_X\omega = 0$.*

5.7. **Symplectomorphisms and Hamiltonian flows.**

6. Constructing new measure preserving systems from old ones

6.1. **Product maps.** Let $(X_1, \mathscr{B}_1, \mu_1)$ and $(X_2, \mathscr{B}_2, \mu_2)$ be two σ-finite measure spaces. The **product σ-algebra** $\mathscr{B}_1 \otimes \mathscr{B}_2$ on $X_1 \times X_2$ is defined as the σ-algebra generated by the family of *rectangles* $\{E_1 \times E_2 : E_i \in \mathscr{B}_i,\ i \in \{1,2\}\}$. Then, one shows that there exists a unique measure $\mu_1 \otimes \mu_2$ on the measurable space $(X_1 \times X_2, \mathscr{B}_1 \otimes \mathscr{B}_2)$, called the *product measure*, such that

$$\mu_1 \otimes \mu_2(E_1 \times E_2) = \mu_1(E_1)\mu_2(E_2), \quad \forall E_1 \in \mathscr{B}_1,\ \forall E_2 \in \mathscr{B}_2.$$

When $f_i\colon (X_1, \mathscr{B}_i, \mu_i) \circlearrowleft$, for $i \in \{1,2\}$, are measure preserving maps, then invoking Theorem 5.1 one can easily show that the measure $\mu_1 \otimes \mu_2$ is $f_1 \times f_2$-invariant, where $f_1 \times f_2\colon X_1 \times X_2 \ni (x_1, x_2) \mapsto \big(f_1(x_1), f_2(x_2)\big) \in X_1 \times X_2$.

By induction, this construction can be easily extended from two to any (finite) number of factor maps. Moreover, this construction can be extended to countably many factors, assuming that each factor is a probability space (see [VO16, Appendix A.2] for details).

6.2. **Induction or renormalization.** Let $(X, \mathscr{B}, \mu)$ be a probability space, $f\colon X \circlearrowleft$ be a measure preserving map and $A \in \mathscr{B}$ a measurable set with $\mu(A) > 0$. By Poincaré recurrence theorem (Corollary 4.4) we know that μ-almost every point $x \in A$ returns to A infinitely many times. So we can define the function $\rho_A\colon A \to \mathbb{N}$ by

$$\rho_A(x) := \min\{n \in \mathbb{N} : f^n(x) \in A\},$$

and this function is well-defined μ-a.e. on A. Then we define the **first return map to** A by

$$f_A(x) := f^{\rho_A(x)}(x),$$

which is well define μ-a.e. on A. In order to see that f_A is a well defined dynamical system, is enough to observe that if we define

$$A_f := A \cap \left(\bigcap_{m\geqslant 0} \bigcup_{n\geqslant m} f^{-n}(A) \right),$$

then by Corollary 4.4 one have that $\mu(A_f) = \mu(A) > 0$, with $A_f \subset A$, and one can easily check that $\rho_A(x) < \infty$ and $f_A(x) \in A_f$, for every $x \in A_f$. So, this means that $f_A \colon A_f \circlearrowleft$ is a well defined dynamical system. Moreover, if define the probability measure μ_A on the measure space $(A, \mathcal{B}_A)$ by

$$\mu_A(E) := \frac{\mu(E)}{\mu(A)}, \quad \forall E \in \mathscr{B}_A,$$

and where $\mathscr{B}_A := \{B \in \mathscr{B} : B \subset A_f\}$, we have the following

Theorem 6.1. *The map $f_A \colon A_f \circlearrowleft$ is a $\mathscr{B}_A$-measurable and the measure μ_A is f_A-invariant.*

Proof. For the sake of simplicity of the notation, from now on we shall assume that $A = A_f$, and for each $n \in \mathbb{N}$, let us consider the set $A_n := \rho^{-1}(n) \subset A$. Observe that

$$A = \bigsqcup_{n\in\mathbb{N}} A_n,$$

and for each $E \subset A$, it holds

$$f_A^{-1}(E) = \bigsqcup_{n\in\mathbb{N}} f_A^{-1}(E) \cap A_n = \bigsqcup_{n\in\mathbb{N}} f^{-n}(E) \cap A_n. \tag{5}$$

On the other hand, one can easily check that

$$A_n = \rho^{-1}(n) = A \cap \left(\bigcap_{i=1}^{n-1} f^{-i}(X\backslash A) \right) \cap f^{-n}(A), \quad \forall n \in \mathbb{N}. \tag{6}$$

From (6) we conclude ρ is a measurable function and then, by (5) we get that f_A is a $\mathscr{B}_A$-measurable map.

Firnally, in order to show that μ_A is f_A-invariant, it is enough to show that $\mu(E) = \mu(f_A^{-1}(E))$, for any $E \in \mathscr{B}_A$. In order to prova that, let us fix a measurable set $E \in \mathscr{B}_A$ and observe that, by (5) we get

$$\mu\big(f_A^{-1}(E)\big) = \sum_{n=1}^{\infty} \mu\big(f^{-n}(E) \cap A_n\big).$$

Let us prove that the right hand side of this equality is equal to $\mu(E)$. To do that, let us start noticing that

$$\begin{aligned} \mu(E) = \mu\big(f^{-1}(E)\big) &= \mu\big(f^{-1}(E) \cap A\big) + \mu\big(f^{-1}(E)\backslash A\big) \\ &= \mu\big(f^{-1}(E) \cap A_1\big) + \mu\big(f^{-1}(E)\backslash A\big), \end{aligned} \tag{7}$$

where the first equality follows from the fact $f^{-1}(E) = \big(f^{-1}(E) \cap A\big) \sqcup \big(f^{-1}(E)\backslash A\big)$, and the second one from $f^{-1}(E) \cap A = f^{-1}(E) \cap A_1$.

Then, focusing on the last term of (7) we get

$$\begin{aligned} \mu\big(f^{-1}(E)\backslash A\big) &= \mu\big(f^{-2}(E)\backslash f^{-1}(A)\big) \\ &= \mu\Big(\big(f^{-2}(E)\backslash f^{-1}(A)\big) \cap A\Big) + \mu\Big(\big(f^{-2}(E)\backslash f^{-1}(A)\big)\backslash A\Big) \\ &= \mu\big(f^{-2}(E) \cap A_2\big) + \mu\Big(f^{-2}(E)\backslash\big(A \cap f^{-1}(A)\big)\Big), \end{aligned} \tag{8}$$

where the first equality follows from the invariance of μ by f, the second one by the partition argument we used above to prove (7), and the last equality follows from the fact that $\big(f^{-2}(E)\backslash f^{-1}(A)\big)\cap A = f^{-1}(E)\cap A_2$.

Combining equations (7) and (8) and repeating inductively the last argument, one can show that

$$\mu(E) = \sum_{i=1}^{n} \mu\big(f^{-i}(E)\cap A_i\big) + \mu\Big(f^{-n}(E)\backslash\big(A\cap f^{-1}(A)\cap\cdots\cap f^{-(n-1)}(A)\big)\Big),$$

for every $n \geqslant 1$.

So, in order to prove that $\mu(E) = \mu\big(f^{-1}(E)\big)$ it is enough to show that

$$\mu\Big(f^{-n}(E)\backslash\Big(\bigcap_{i=1}^{n-1} f^{-i}(A)\Big)\Big) \leqslant \mu\Big(f^{-n}(A)\backslash\Big(\bigcup_{i=1}^{n-1} f^{-i}(A)\Big)\Big) \to 0$$

as $n \to +\infty$. But this easily follows from the fact that $f^{-1}(A)\backslash A, f^{-2}(A)\backslash(A\cup f^{-1}(A))\ldots$ are pairwise disjoint and hence,

$$1 = \mu(X) \geqslant \sum_{n=1}^{\infty} \mu\Big(f^{-n}(A)\backslash\Big(\bigcup_{i=1}^{n-1} f^{-i}(A)\Big)\Big).$$

□

6.3. **Kakutani-Rokhlin towers.** In this section we shall try to revert the previous construction we performed in §6.2. To do that, for the time being let $(X,\mathscr{B},\mu)$ be an arbitrary measure space, $\rho\colon X\to\mathbb{N}_0$ be an integrable function and $f\colon X\circlearrowleft$ be a measure preserving map.

We first onsider the new measure space $(X_\rho,\mathscr{B}_\rho,\mu_\rho)$ where each object is defined as follows: we first define the space X_ρ by

$$X_\rho := \bigsqcup_{n\in\mathbb{N}_0}\bigsqcup_{0\leqslant i\leqslant n} \rho^{-1}(n)\times\{i\} \subset X\times\mathbb{N}_0;$$

the σ-algebra $\mathscr{B}_\rho$ is defined as the one generated by the family

$$\mathscr{B}_\rho := \sigma\Big\{A\times\{i\}\subset X\times\mathbb{N}_0 : A\in\mathscr{B},\ \exists n\in\mathbb{N}_0,\ 0\leqslant i\leqslant n,\ A\subset\rho^{-1}(n)\Big\};$$

and the measure μ_ρ is characterized as the only measure on the measurable space $(X_\rho,\mathscr{B}_\rho)$ such that

$$\mu_\rho(A\times\{i\}) := \left(\int_X \rho+1\,\mathrm{d}\mu\right)^{-1}\mu(A),$$

for every generating set $A\times\{i\}$ of $\mathscr{B}_\rho$.

Exercise 12. *Show that $(X_\rho,\mathscr{B}_\rho,\mu_\rho)$ is a well-defined probability space.*

Then we define the measure preserving map $f_\rho\colon X_\rho\circlearrowleft$ by

$$f_\rho(x,i) := \begin{cases}(x,i+1), & \text{for } x\in X,\ 0\leqslant i<\rho(x);\\ (f(x),0), & \text{for } x\in X,\ i=\rho(x).\end{cases}$$

Exercise 13. *Show that the map $f_\rho\colon(X_\rho,\mathscr{B}_\rho,\mu_\rho)\circlearrowleft$ is indeed a measure preserving map. Now consider the first return map (also called induction as in §6.2) of the map f_ρ to the set $A := X\times\{0\}$. Show that $\mu_\rho(A)>0$ and the first return map is "isomorphic" to the original $f\colon(X,\mathscr{B},\mu)\circlearrowleft$. Can you give a definition of* isomorphism *between measure preserving maps?*

7. Existence of invariant measures for continuous maps

The main porpuse of this section consists in proving the following

Theorem 7.1. *Let X be compact metric space and $f\colon X \circlearrowleft$ be a continuous map. Then, there exists an f-invariant Borel probability measure.*

In order to prove Theorem 7.1 we first need to introduce some concepts in Functional Analysis.

7.1. **Linear operators and dual spaces.** Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be two Banach spaces over the field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. A linear map $A\colon E \to F$ is continuous if and only its **norm**

$$\|A\| := \sup_{x \in X\setminus\{0\}} \frac{\|Ax\|_F}{\|x\|_E} < \infty. \tag{9}$$

The space of linear continuous maps from E to F is a denoted by $\mathcal{L}(E,F)$ and it turns to be a Banach space when its endowed with the norm given by (9).

When F is the field itself $\mathbb{K}$, a continuous linear operator $\eta\colon E \to \mathbb{K}$ is called a continuous linear functional. The **dual space** $E^\star$ is defined as the linear space $\mathcal{L}(E,\mathbb{K})$. Then we can define the norm $\|\cdot\|_{E^\star}$ on $E^\star$ by

$$\|\eta\|_{E^\star} := \sup\{|\eta(x)| : x \in E,\ \|x\|_E = 1\}, \quad \forall \eta \in E^\star.$$

The linear space $E^\star$ turns into a Banach space when it is endowed with the norm $\|\cdot\|_{E^\star}$. The topology on $E^\star$ induced by the norm $\|\cdot\|_{E^\star}$ is usually called the **strong topology**.

On the other hand, any continuous linear operator $A\colon E \to F$ naturally induces the so callled **adjoint operator** $A\colon F^\star \to E^\star$ given by

$$A^\star\eta(x) := \eta\big(A(x)\big), \quad \forall x \in E,\ \forall \eta \in F^\star.$$

Exercise 14. *Show that the adjoint operator is indeed a linear continuous operator and $\|A\| \geqslant \|A^\star\|$.*

7.2. **Weak topologies.** Given a topological space (Y, τ_Y), a non-empty set X and an arbitrary family of maps $\mathscr{F} = \{f_i\colon X \to Y\}_{i\in I}$, we define the **weak topology** on X induced by τ_Y and $\mathscr{F}$ as the weakest (i.e. the coarsest) topology τ on X such that $f_i\colon (X,\tau) \to (Y,\tau_Y)$ is continuous, for every $i \in I$.

Now, if $(E, \|\cdot\|_E)$ denotes a Banach space over $\mathbb{K} = \mathbb{R}, \mathbb{C}$, and $E^\star$ is its dual space, we can define the so called **weak topology** on E as the weak topology induced by the family $E^\star$, i.e. it is the weakest topology on E such that every $\eta \in E^\star$ is continuous on E. In general, the weak topolgy is weaker than the Banach topology induced by the norm $\|\cdot\|_E$.

Exercise 15. *Show that if E is a Banach space, then norm and the weak topologies coincide if and only if E has finite dimension.*

Now, we can consider the space $E^{\star\star} := (E^\star)^\star$ which is the dual space of the Banach space $(E^\star, \|\cdot\|_{E^\star})$. Analogously, we can endow the space $E^\star$ with the **weak topology** which is the weakest topolgy such that every element of $E^{\star\star}$ is continuous.

However, there shall need to endow the space $E^\star$ with a weaker topology which is defined as follows. One can show that there is a natural injection of E into $E^{\star\star}$, i.e. a linear continuous injective map $j\colon E \to E^{\star\star}$ which is given by

$$j(x)(\eta) := \eta(x), \quad \forall x \in E,\ \forall \eta \in E^\star.$$

Then, the so called **weak-$\star$ topology** is the weakest topolgy on $E^\star$ such that every linear functional in $j(E)$ is continuous. Given a sequence $(\phi_n)_{n\geqslant 1} \subset E^\star$, we shall write

$$\eta_n \rightharpoonup \eta \in E^\star, \quad \text{as } n \to \infty,$$

when the sequence $(\eta_n)_{n\geqslant 1}$ converges to η in the weak-$\star$ topology.

Exercise 16. *Show that the weak and weak-$\star$ topologies coincide on $E^\star$ if and only if E has finite dimension. On the other hand, show that that the* closed unit ball *on a Banach space E, which is given by $\{x \in E : \|x\|_E \leqslant 1\}$ is compact if and only if E has finite dimension.*

We shall need the following

Theorem 7.2 (Banach-Alaoglu theorem)**.** *The closed unit ball $\{\eta \in E^\star : \|\eta\|_{E^\star} \leqslant 1\}$ is a compact set when $E^\star$ is with its weak-$\star$ topology.*

Proof. See for instance Einsiedler and Ward [EW17, Theorem 8.10]. □

7.3. **The space of continuous functions.** Let (X, d) be a compact metric space and write $C^0(X)$ to denote the space of continuous real functions on X. Since X is compact, any continuous function is bounded and hence, we can define the norm

$$\|\phi\|_0 := \sup_{x\in X} |\phi(x)|, \quad \forall \phi \in C^0(X).$$

Exercise 17. *Show that $C^0(X)$ is a Banach space when it is endowed with the norm $\|\cdot\|$.*

Given an arbitrary continuous map $f\colon X \circlearrowleft$, we can consider the **induced pull-back linear map,** also called the **Koopman operator** $U_f\colon C^0(X) \circlearrowleft$ given by

$$U_f(\phi) := \phi \circ f, \quad \forall \phi \in C^0(X).$$

Exercise 18. *Show that U_f is a continuous linear operator and its norm*

$$\left\|U_f\right\| := \sup\left\{\left\|U_f(\phi)\right\|_0 : \phi \in C^0(X), \|\phi\|_0 = 1\right\} = 1.$$

Moreover, $\left\|U_f\phi\right\| = \|\phi\|$, for every $\phi \in C^0(X)$ if and only if f is surjective.

On the other hand, we have the following

Theorem 7.3 (Riesz representation theorem)**.** *If $\eta\colon C^0(X) \to \mathbb{R}$ is a continuous linear functional on the Banach space $(C^0(X), \|\cdot\|)$, then there exist two finite Borel measures $\mu^=$ and $\mu^=$ on X such that*

$$\eta(\phi) = \int_X \phi \, \mathrm{d}\mu^+ - \int_X \phi \, \mathrm{d}\mu^-, \quad \forall \phi \in C^0(X).$$

Proof. See for instance Einsiedler and Ward [EW17, Theorem 7.44]. □

Then, let us write $\mathfrak{M}(X)$ for the space of Borel probability measures on X. By Theorem 7.3 we can consider $\mathfrak{M}(X)$ as a subset of unit ball of the dual space $C^0(X)^\star$. On the other hand, by Exercise 3 we know that the restriction of the adjoint operator of U_f to the space $\mathfrak{M}(X)$ coincide with the push-forward operator $f_{star}\colon \mathfrak{M}(X) \circlearrowleft$. In particular, we conclude $f_\star\colon \mathfrak{M}(X) \circlearrowleft$ is continuous when $\mathfrak{M}(X)$ is endowed with the weak-$\star$ topology that inherits from the inclusion $\mathfrak{M}(X) \subset C^0(X)^\star$.

Exercise 19. *Show that the space of Borel probability measures $\mathfrak{M}(X)$ is a closed subset of the closed unit ball $\bar{B} := \left\{\xi \in C^0(X)^\star : \|\xi\| \leqslant 1\right\}$ when it is endowed with the weak-$\star$ topology. Then, invoking Theorem 7.2, conclude that $\mathfrak{M}(X)$ is compact with the weak-$\star$ topology.*

Now we can complete the

Proof of Theorem 7.1. Let ν be any Borel probability measure on X. For each $n \in \mathbb{N}$, let us consider the measure $\mu_n \in \mathfrak{M}(X)$ given by

$$\mu_n := \frac{1}{n} \sum_{j=0}^{n-1} f_\star^j \nu.$$

By Exercise 19 we know that $\mathfrak{M}(X)$ is compact for the weak-$\star$ topology. So, there exists a subsequent $(\mu_{n_j})_{j \geqslant 1}$ and $\mu \in \mathfrak{M}(X)$ such that $\mu_{n_j} \rightharpoonup \mu$, as $n_j \to \infty$. The, since $f_\star \colon \mathfrak{M}(X) \circlearrowleft$ is continuous, we get $f_\star \mu_{n_j} \rightharpoonup f_\star \mu$, as $n_j \to \infty$. So, in order to show that μ is f-invariant, it is enough to show that $f_\star \mu_n - \mu_n \rightharpoonup 0$, as $n \to \infty$. In order to do that, let $\phi \in C^0(X)$ be arbitrary and notice that

$$\int_x \phi \, \mathrm{d}(f_\star \mu_n - \mu_n) = \frac{1}{n} \sum_{j=0}^{n-1} \left(\int_X \phi \, \mathrm{d} f_\star^{j+1} \nu - \int_X \phi \, \mathrm{d} f_\star^j \nu \right) = \frac{1}{n} \int_X \phi \circ f^n - \phi \, \mathrm{d}\nu \to 0,$$

as $n \to +\infty$. □

Exercise 20. *Let X and Y be two compact metric spaces, and $f \colon X \circlearrowleft$ and $g \colon X \times Y \to Y$ two continuous functions. Let us consider the so called* skew-product map *$H \colon X \times Y \circlearrowleft$ given by*

$$H(x,y) := \big(f(x), g(x,y)\big), \quad \forall (x,y) \in X \times Y.$$

Given any invariant measure $\mu \in \mathfrak{M}(f)$, let us show that there exists an invariant measure $\hat{\mu} \in \mathfrak{M}(H)$ such that $\pi_\star \hat{\mu} = \mu$, where $\pi \colon X \times Y \to X$ denotes the projection on the first coordinate.

8. Ergodic theorems

Let $(X, \mathscr{B}, \mu)$ be a probability space and $f \colon X \circlearrowleft$ a measure preserving endomorphism. Let $E \in \mathscr{B}$ be a subset such that $\mu(E) > 0$. Then by Poincare's recurrence theorem (Corollary 4.4) we know that the set $\{i \in \mathbb{N}_0 : f^i(x) \in E\}$ is infinite for μ-almost every point $x \in E$. This can be considered as a qualitative information. In this section we shall focus on the problem of finding quantitative estimates for the frequency of sojourn

$$\frac{1}{n} \sharp \{i \in \mathbb{N}_0 : i < n, \ f^i(x) \in E\} = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{1}_E \circ f^i(x),$$

as $n \to \infty$, and where $\mathbb{1}_E \colon X \to \{0,1\}$ denotes the indicator function of the set E. To do that, we shall analyze the asymptotic behavior of so called **Birkhoff averages** as the following one:

$$\mathcal{S}_f^n \phi := \frac{1}{n} \sum_{j=0}^{n-1} \phi \circ f^j,$$

for an arbitrary integrable function $\phi \colon X \to \mathbb{R}$.

The first result concerning the asymptotic behavior of Birkhoff averages is the celebrated

Theorem 8.1 (von Neumann Ergodic Theorem)**.** *Let $(x, \mathscr{B}, \mu)$ be a probability space and $f \colon X \circlearrowleft$ be a measure preserving endomorphism. Then, for every $\phi \in L^2(X, \mathscr{B}, \mu)$, there is a $\tilde{\phi} \in L^2(X, \mathscr{B}, \mu)$ such that*

$$\frac{1}{n} \sum_{j=0}^{n-1} \phi \circ f^j \to \tilde{\phi}, \quad \text{in } L^2(X, \mathscr{B}, \mu),$$

as $n \to +\infty$. Moreover, it holds

$$\tilde{\phi}(x) = \tilde{\phi}\big(f(x)\big), \quad \text{for } \mu\text{-a.e. } x \in X.$$

In order to prove Theorem 8.1, we first need to revise some concepts on Lebesgue and Hilbert spaces.

8.1. **Revision on Lebesgue spaces.** Let $(X, \mathcal{B}, \mu)$ be an arbitrary measure space. For each real number p with $1 \leqslant p < \infty$, let us consider the space $L^p(X, \mathcal{B}, \mu)$ of (equivalence classes of) measurable functions $\phi\colon X \to \mathbb{C}$ such that $|\phi|^p$ is integrable, and where two functions are equivalent if and only if they coincide μ-almost everywhere on X. Then one can show that the $L^p(X, \mathscr{B}, \mu)$ is a Banach space when it is endowed with the norm

$$\|\phi\|_p := \left(\int_X |\phi|^p \, \mathrm{d}\mu\right)^{1/p}, \quad \forall \phi \in L^p(X, \mathscr{B}, \mu).$$

On the other hand, we define the space $L^\infty(X, B, \mu)$ as the space of (equivalence classes of) measurable functions $\phi\colon X \to \mathbb{C}$ such that there exists $C > 0$ satisfying $|\phi(x)| \leqslant C$, for μ-almost every $x \in X$, and where two functions are equivalent when they coincide μ-almost everywhere. The space $L^\infty(X, \mathscr{B}, \mu)$ is a Banach space when it is endowed with the norm

$$\|\phi\|_\infty := \inf\{C > 0 : |\phi(x)| \leqslant C, \text{ for } \mu\text{-a.e. } x \in X\}, \quad \forall \phi \in L^\infty(X, \mathscr{B}, \mu).$$

Exercise 21. *If $(X, \mathscr{B}, \mu)$ is a probability space, $1 \leqslant p < q \leqslant \infty$ and $\phi \in L^q(X, \mathscr{B}, \mu)$, then show that $\phi \in L^p(X, \mathscr{B}, \mu)$ and $\|\phi\|_p \leqslant \|\phi\|_q$. On the other hand, show that*

$$\|\psi\|_\infty = \lim_{p \to \infty} \|\psi\|_p, \quad \forall \psi \in L^\infty(X, \mathscr{B}, \mu).$$

Exercise 22. *Consider the map $\langle \cdot, \cdot \rangle\colon L^2(X, \mathscr{B}, \mu) \times L^2(X, \mathscr{B}, \mu) \to \mathbb{C}$ given by*

$$\langle \phi, \psi \rangle := \int_X \phi \overline{\psi} \, \mathrm{d}\mu, \quad \forall \phi, \psi \in L^2(X, \mathscr{B}, \mu).$$

Show that $\langle \cdot, \cdot \rangle$ is well defined, is a scalar product and it turns $L^2(X, \mathscr{B}, \mu)$ into a Hilbert space.

A Banach space is said to be Hibertlizable *when one can define a scalar product on it that turns it into a Hilbert space and whose topology coincides with the original one given by the Banach structure.*

Show that the Lebesgues space $L^p(X, \mathscr{B}, \mu)$ is Hilbertlizable if and only if $p = 2$.

Given a probability space $(X, \mathscr{B}, \mu)$, a measure preserving map $f\colon X \circlearrowleft$, and a measurbale function $\phi\colon X \to \mathbb{C}$, consider the new measurable function $U_f\phi := \phi \circ f$. The map U_f is called the **Koopman operator**.

Exercise 23. *For every $p \in [1, \infty]$, show that $U_f\phi \in L^p(X, \mathscr{B}, \mu)$, whenever $\phi \in L^p(X, \mathscr{B}, \mu)$. Moreover, show that $U_f\colon L^p(X, \mathscr{B}, \mu) \circlearrowleft$ is a linear isometry, i.e. $\left\|U_f\phi\right\|_p = \|\phi\|_p$, for every $\phi \in L^p(X, \mathscr{B}, \mu)$ and every $p \in [1, \infty]$.*

8.2. **Revision of Hilbert space theory.** Let $(\mathscr{H}, \langle \cdot, \cdot \rangle)$ be a Hilbert over the field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. We will write $E < \mathscr{H}$ when E is a closed linear subspace. Given any non-empty subset $F \subset \mathscr{H}$, we define its **orthogonal complement** $F^\perp$ by

$$F^\perp := \{x \in \mathscr{H} : \langle x, y \rangle = 0, \ \forall y \in F\}.$$

Observe that $F^\perp < \mathscr{H}$ and $F \cap F^\perp$ is either empty or the singleton $\{0\}$.

We will need the following

Theorem 8.2 (Projection on convex subsets)**.** *Let $K \subset \mathscr{H}$ be a non-empty convex closed set. Then there is a continuous map $P_K \colon \mathscr{H} \to K$ such that for every $x \in \mathscr{H}$ the vector $P_K(x)$ is the only one in K such that*

$$\|x - P_k(x)\| = \inf\{\|x - y\| : y \in K\}.$$

Moreover, if K is closed linear subspace, then P_K is a linear continuous map such that

$$\langle x - P_K(x), x\rangle = 0, \quad \forall x \in \mathscr{H}.$$

In particular, it holds $\ker P_K = K^\perp$ and $K \oplus K^\perp = \mathscr{H}$.

Proof. See for instance [EW17, Theorem 3.13 and Corollary 3.17]. □

Let $\mathscr{H}^\star$ denote its dual space, i.e. the space of continuous linear functionals on $\mathscr{H}$. We have the following

Theorem 8.3 (Riesz representation theorem for Hilbert spaces)**.** *For every $\eta \in \mathscr{H}^\star$, there exists a unique $v_\eta \in \mathscr{H}$ such that*

$$\eta(x) = \langle x, v_\eta\rangle, \quad \forall x \in \mathscr{H}.$$

Proof. See for instance [EW17, Corollary 3.19]. □

Theorem 8.3 lets us redefine the concept of adjoint operator on Hilbert spaces. Given two Hilbert spaces $(\mathscr{H}, \langle\cdot,\cdot\rangle_{\mathscr{H}})$ and $(\mathscr{J}, \langle\cdot,\cdot\rangle_{\mathscr{J}})$ over $\mathbb{K} = \mathbb{R}, \mathbb{C}$, and a continuous linear operator $A \in \mathcal{L}(\mathscr{H}, \mathscr{J})$, we define its **adjoint operator** $A^\star \colon \mathscr{J} \to \mathscr{H}$ by the following property:

$$\langle Ax, y\rangle = \langle x, A^\star y\rangle, \quad \forall x \in \mathscr{H},\ \forall y \in \mathscr{J}.$$

Exercise 24. *Show that $A^\star \in \mathcal{L}(\mathscr{J}, \mathscr{H})$, for every $A \in \mathcal{L}(\mathscr{H}, \mathscr{J})$.*

A linear operator $U \in \mathcal{L}(\mathscr{H}, \mathscr{J})$ is said to be an **isometry** when

$$\|Ux\|_{\mathscr{J}} = \|x\|_{\mathscr{H}}, \quad \forall x \in \mathscr{H}.$$

Exercise 25. *Let $U \colon \mathscr{H} \to \mathscr{J}$ be a continuous linear operator. Show that the following statements are equivalent:*

(1) U is an isometry;
(2) $\langle Ux, Uy\rangle_{\mathscr{J}} = \langle x, y\langle_{\mathscr{H}}$, for every $x, y \in \mathscr{H}$;
(3) $U^\star U = I$, where $I \in \mathcal{L}(\mathscr{H})$ denotes the identity map.

A surjective isometry $U \in \mathcal{L}(\mathscr{H})$ is called a **unitary operator**. One can easily check that U is a unitary operator if and only if $U^\star U = UU^\star = I$.

Now we can prove the so called von Neumann's ergodic theorem for isometries on Hilbert spacees:

Theorem 8.4. *Let $\mathscr{H}$ be a Hilbert space over $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and $U \colon \mathscr{H} \to \mathscr{H}$ be a linear isometry. Let $\mathscr{I} := \ker(U - I) < \mathscr{H}$ and $P_{\mathscr{I}} \colon \mathscr{H} \to \mathscr{I}$ the projection map given by Theorem 8.2. Then, it holds*

$$\frac{1}{n}\sum_{i=0}^{n-1} U^i x \to P_{\mathscr{I}}(x), \quad \forall x \in \mathscr{H},$$

as $n \to +\infty$.

Proof. Let us start considering the closed linear space

$$E = \overline{(U - I)\mathscr{H}}.$$

Let us show that $\mathscr{I} = E^\perp$. In fact, first observe that for every $x \in \mathscr{H}$ and any $y \in \mathscr{I}$ we have

$$\langle Ux - x, y\rangle = \langle Ux, y\rangle - \langle x, y\rangle = \langle Ux, Uy\rangle - \langle x, y\rangle = 0,$$

where the last equality follows from the fact that U is an isometry and Exercise 25. So, $\mathscr{I} \subset E^{\perp}$.

On the other hand, if $y \in E^{\perp}$, then we have

$$0 = \langle Ux - x, y\rangle = \langle Ux, y\rangle - \langle x, y\rangle = \langle x, U^{\star}y\rangle - \langle x, y\rangle = \langle x, U^{\star}y - y\rangle,$$

for every $x \in \mathscr{H}$. So, by Theorem 8.3 we have that $U^{\star}y - y = 0$. Then we have

$$\begin{aligned}\|Uy - y\|^2 &= \langle Uy - y, Uy - y\rangle = \|Uy\|^2 - \langle Uy, y\rangle - \langle y, Uy\rangle + \|y\|^2\\ &= \|y\|^2 - \langle y, U^{\star}y\rangle - \langle U^{\star}y, y\rangle - \|y\|^2\\ &= \|y\|^2 - \|y\|^2 - \|y\|^2 + \|y\|^2 = 0.\end{aligned}$$

So, $y \in \mathscr{I}$ and we have proved that $\mathscr{I} = E^{\perp}$. Hence, it holds $\mathscr{H} = \mathscr{I} \oplus E$ and in order to conclude the proof it is enough to show that

$$\frac{1}{n}\sum_{i=0}^{n-1} U^i x \to 0, \quad \text{as } n \to +\infty, \tag{10}$$

for every $x \in E$. To do that, let x denote an arbitrary point of E and $\varepsilon > 0$ be any positive real number. So, there exists a $y \in \mathscr{H}$ such that

$$\|x - (Uy - y)\| < \frac{\varepsilon}{2},$$

and we get

$$\begin{aligned}\left\|\frac{1}{n}\sum_{i=0}^{n-1} U^i x\right\| &= \frac{1}{n}\left\|\sum_{i=0}^{n-1} U^i\Big((Uy - y) + x - (Uy - y)\Big)\right\|\\ &\leqslant \frac{1}{n}\left\|\sum_{i=0}^{n-1} U^{i+1}y - U^i y\right\| + \frac{1}{n}\sum_{i=0}^{n-1}\left\|U^i\big(x - (Uy - y)\big)\right\|\\ &= \frac{\|U^n y - y\|}{n} + \frac{1}{n}\sum_{i=0}^{n-1}\|x - (Uy - y)\| \leqslant \frac{2}{n}\|y\| + \frac{\varepsilon}{2} \leqslant \varepsilon,\end{aligned}$$

for every $n \geqslant \dfrac{4\|y\|}{\varepsilon}$. Since $\varepsilon > 0$ is arbitrary, we conclude that (10) holds. □

Proof of Theorem 8.1. It is a straightforward consequence of Theorem 8.4 and the fact that the Koopman operator $U_f \colon L^2(X, \mathscr{B}, \mu) \circlearrowleft$ is an isometry (Exercise 23). □

Here we review some interesting consequences of von Neumann's ergodic theorem (Theorem 8.1):

Corollary 8.5. *Let $(X, \mathscr{B}, \mu)$ be a probability space and $f \colon X \circlearrowleft$ be a measure preserving automorphism. Then, for every $\phi \in L^2(X, \mathscr{B}, \mu)$, there exists $\tilde{\phi} \in L^2(X, \mathscr{B}, \mu)$ such that*

$$\lim_{n\to+\infty}\frac{1}{n}\sum_{j=0}^{n-1}\phi \circ f^i = \lim_{n\to-\infty}\frac{1}{n}\sum_{i=0}^{n-1}\phi \circ f^{-i} = \tilde{\phi},$$

Where both limits are taken in L^2.

Proof. This is a straightforward consequence of Theorem 8.4 and the simple fact that U_f and $U_{f^{-1}} = {U_f}^{-1}$ are isometries, and $\ker(U_f - I) = \ker(U_{f^{-1}} - I)$ □

Corollary 8.6. *Let $(X, \mathscr{B}, \mu)$ be a probability space, $f\colon X \circlearrowleft$ be measure preserving map and $\phi \in L^1(X, \mathscr{B}, \mu)$. Then, there is $\tilde{\phi} \in L^1(X, \mathscr{B}, \mu)$ such that*

$$\frac{1}{n}\sum_{i=0}^{n-1} \phi \circ f^i \to \tilde{\phi}, \quad \text{in } L^1,$$

as $n \to +\infty$.

Proof. For the sake of simplicity of notation, given any $\psi\colon X \to \mathbb{R}$ and any $n \in \mathbb{N}$, let us write $\mathcal{S}_f^n \psi$ for the Birkhoff average of order n of ψ over f, i.e. we have

$$\mathcal{S}_f^n \psi = \frac{1}{n}\sum_{j=0}^{n-1} \psi \circ f^j.$$

Let us fix a real number $\varepsilon > 0$. Since $L^2(X, \mathscr{B}, \mu)$ is dense in $L^1(X, \mathscr{B}, \mu)$, there is $\xi \in L^2(X, \mathscr{B}, \mu)$ such that

$$\|\phi - \xi\|_1 < \frac{\varepsilon}{4},$$

and then,

$$\left\|\mathcal{S}_f^n \phi - \mathcal{S}_f^n \xi\right\|_1 < \frac{\varepsilon}{4}, \quad \forall n \geqslant 1.$$

On the other hand, by Theorem 8.1 we know that there is $\tilde{\xi} \in L^2(X, \mathscr{B}, \mu)$ such that

$$\left\|\mathcal{S}_f^n \xi - \tilde{\xi}\right\|_2 \to 0, \quad \text{as } n \to +\infty,$$

and thus, there is $N \in \mathbb{N}$ such that

$$\left\|\mathcal{S}_f^n \xi - \tilde{\xi}\right\|_1 \leqslant \left\|\mathcal{S}_f^n \xi - \tilde{\xi}\right\|_2 < \varepsilon/4, \quad \forall n \geqslant N.$$

Putting together all these estimates we get

$$\left\|\mathcal{S}_f^m \phi - \mathcal{S}_f^n \phi\right\|_1 \leqslant \left\|\mathcal{S}_f^m \phi - \mathcal{S}_f^m \xi\right\|_1 + \left\|\mathcal{S}_f^m \xi - \tilde{\xi}\right\|_1 + \left\|\tilde{\xi} - \mathcal{S}_f^n \xi\right\|_1 + \left\|\mathcal{S}_f^n \xi - \mathcal{S}_f^n \phi\right\|_1 < \varepsilon,$$

for every $m, n \geqslant N$. So, $\left(\mathcal{S}_f^n \phi\right)_{n \geqslant 1}$ is a Cauchy sequence in $L^1(X, \mathscr{B}, \mu)$ and hence, there exists its limit $\tilde{\phi} \in L^1(X, \mathscr{B}, \mu)$. □

Now we are going to prove that indeed pointwise convergence holds on Corollary 8.6:

Theorem 8.7 (Birkhoff ergodic theorem)**.** *Let $(X, \mathscr{B}, \mu)$ be a probability space, $f\colon X \circlearrowleft$ a measure preserving map and $\phi\colon X \to \mathbb{R}$ any integrable function. Then there is $\tilde{\phi} \in L^1(X, \mathscr{B}, \mu)$ such that*

$$\mathcal{S}_f^n \phi(x) = \frac{1}{n}\sum_{j=0}^{n-1} \phi\big(f^j(x)\big) \to \tilde{\phi}(x), \quad \text{as } n \to \infty,$$

for μ-almost every $x \in X$.

Moreover, it holds

$$\int_X \phi \, \mathrm{d}\mu = \int_X \tilde{\phi} \, \mathrm{d}\mu,$$

and

$$\tilde{\phi}(x) = \tilde{\phi}\big(f(x)\big), \quad \text{for } \mu\text{-a.e. } x \in X.$$

In order to prove Theorem 8.7 we first need some preelimiary results:

Lemma 8.8 (Borel-Cantelli)**.** *Let $(X, \mathscr{B}, \mu)$ be a probability space and $A_1, A_2, \ldots \in \mathscr{B}$ be a sequence of measurable sets. If*

$$\sum_{n\geqslant 1} \mu(A_n) < \infty,$$

then it holds

$$\mu\Big(\limsup_n A_n\Big) = 0,$$

where the limit superior *of the sequence (A_n) is defined by*

$$\limsup_n A_n := \bigcap_{m\geqslant 1} \bigcup_{n\geqslant m} A_n.$$

Proof. For each $m \in \mathbb{N}$, let us consider the set

$$B_m := \bigcup_{n\geqslant m} A_n.$$

Observe that $B_1 \supset B_2 \supset \cdots$ and since $\sum_{n\geqslant 1} \mu(A_n) < \infty$, it holds $\mu(B_m) \leqslant \sum_{n\geqslant m} \mu(A_n) \to 0$, as $m \to \infty$. Finally, notice that

$$\mu\Big(\limsup_n A_n\Big) = \mu\Big(\bigcap_{m\geqslant 1} B_m\Big) = \lim_{m\to\infty} \mu(B_m) = 0.$$

□

Lemma 8.9. *Let $(X, \mathscr{B}, \mu)$ be a probability space and $\phi \in L^1(X, \mathscr{B}, \mu)$. Then it holds*

$$\lim_{n\to+\infty} \frac{\phi\big(f^n(x)\big)}{n} = 0, \quad \text{for } \mu\text{-a.e. } x \in X.$$

Proof. For the sake of simplicity of notation, given any $\psi\colon X \to \mathbb{R}$ and any $A \subset \mathbb{R}$ and any $r \in \mathbb{R}$ let us write

$$[\psi \in A] := \psi^{-1}(A), \quad \text{and} \quad [\psi > r] := [\psi \in (r, \infty)].$$

Then, first observe that given an arbitrary point $x \in X$, the sequence $\phi(f^n(x))/n$ does not converge to 0 if and only if the limit superior of $|\phi(f^n(x))|/n$ is positive. So, we have

$$\left\{ x \in X : \frac{\phi\big(f^n(x)\big)}{n} \nrightarrow 0, \text{ as } n \to +\infty \right\} = \bigcup_{p\geqslant 1} \bigcap_{m\geqslant 1} \bigcup_{n\geqslant m} \left[\frac{|\phi \circ f^n|}{n} > \frac{1}{p} \right].$$

Thus in order to prove this lemma, it is enough to show that given any $\varepsilon > 0$,

$$\mu\left(\limsup_n \left[\frac{|\phi \circ f^n|}{n} > \varepsilon \right] \right) = 0,$$

and by Lemma 8.8, to prove this last equality it is enough to show that

$$\sum_{n\geqslant 1} \mu\left[\frac{|\phi \circ f^n|}{n} > \varepsilon \right] < \infty.$$

Finally, to prove the convergence of this series, observe that

$$\begin{aligned}
\sum_{n\geqslant 1} \mu\left[\frac{|\phi \circ f^n|}{n} > \varepsilon \right] = \sum_{n\geqslant 1} \mu\left(f^{-n}\left[\frac{|\phi|}{n} > \varepsilon \right] \right) &= \sum_{n\geqslant 1} \mu\left[\frac{|\phi|}{n} > \varepsilon \right] \\
\sum_{n\geqslant 1} \mu\left[\frac{|\phi|}{\varepsilon} > n \right] &= \sum_{n\geqslant 1} \sum_{j\geqslant n} \mu\left[j < \frac{|\phi|}{\varepsilon} \leqslant j+1 \right] \\
&= \sum_{j\geqslant 1} j\mu\left[j < \frac{|\phi|}{\varepsilon} \leqslant j+1 \right] \leqslant \int_X \frac{|\phi|}{\varepsilon}\, \mathrm{d}\mu < \infty.
\end{aligned}$$

□

Theorem 8.10 (Maximal ergodic theorem)**.** *Let $(X,\mathscr{B},\mu)$, f and ϕ as in Theorem 8.7. For any $\alpha>0$, let us consider the set*

$$E^{\alpha}(\phi):=\left[\sup_{n\geqslant 1}\mathcal{S}_f^n\phi>\alpha\right]=\left\{x\in X:\sup_{n\geqslant 1}\frac{1}{n}\sum_{j=0}^{n-1}\phi\big(f^j(x)\big)>\alpha\right\}.$$

Then it holds

$$\int_{E^{\alpha}(\phi)}\phi\,\mathrm{d}\mu\geqslant\alpha\mu\big(E^{\alpha}(\phi)\big).$$

Proof of Theorem 8.10. Let us start noticing that given any $\alpha\in\mathbb{R}$ and any $\phi\in L^1(X,\mathscr{B},\mu)$, it holds

$$E^{\alpha}(\phi)=E^0(\phi-\alpha),$$

and therefore,

$$\int_{E^0(\phi-\alpha)}\phi-\alpha\,\mathrm{d}\mu\geqslant 0\quad\Longleftrightarrow\quad\int_{E^{\alpha}(\phi)}\phi\,\mathrm{d}\mu\geqslant\alpha\mu(E^{\alpha}(\phi)).$$

So, it is enough to prove this theorem for $\alpha=0$.

For each $n\geqslant 1$, let us consider the function

$$\phi_n(x):=\sup\left\{\sum_{j=0}^{k-1}\phi\big(f^j(x)\big):1\leqslant k\leqslant n\right\},\quad\forall x\in X.$$

If we write $[\phi_n>0]:=\{x\in X:\phi_n(x)>0\}$, so it holds $[\phi_n>0]\subset[\phi_{n+1}>0]$, for every $n\geqslant 1$ and

$$E^0(\phi):=\bigcup_{n\geqslant 1}[\phi_n>0].$$

Thus, to prove that $\int_{E^0(\phi)}\phi\,\mathrm{d}\mu\geqslant 0$, by the dominated convergence theorem it is enough to show that

$$\int_{[\phi_n>0]}\phi\,\mathrm{d}\mu\geqslant 0,\quad\forall n\geqslant 1.$$

Then, given any $n\geqslant 1$ and any $m\in\{1,2,\dots,n\}$, we know that

$$\phi_n(x)\geqslant\sum_{j=0}^{m-1}\phi\big(f^j(x)\big),\quad\forall x\in X,$$

and hence,

$$\phi_n\big(f(x)\big)+\phi(x)\geqslant\sum_{j=0}^{m}\phi\big(f^j(x)\big),\quad\forall x\in X,\ \forall m\in\{1,\dots,n\}.\tag{11}$$

Fixing a point $x\in X$, let us consider the two possible cases: either $\phi_n\big(f(x)\big)\geqslant 0$ or $\phi_n\big(f(x)\big)<0$.

In the first case, when $\phi_n\big(f(x)\big)\geqslant 0$, we have $\phi_n\big(f(x)\big)+\phi(x)\geqslant\phi(x)$. Then, by estimate (11) it follows that

$$\phi_n\big(f(x)\big)+\phi(x)\geqslant\phi_n(x),\quad\forall x\in[\phi_n\circ f\geqslant 0].$$

In the second case, when $\phi_n\big(f(x)\big)<0$, by (11) we get

$$\phi(x)\geqslant\sum_{j=0}^{m}\phi\big(f^j(x)\big)-\phi_n\big(f(x)\big)>\sum_{j=0}^{m}\phi\big(f^j(x)\big),\quad\forall m\in\{1,\dots,n\},$$

and then, $\phi_n(x)=\phi(x)$, whenever $x\in[\phi_n\circ f<0]$.

Putting together these estimates we get

$$\begin{aligned}
\int_{[\phi_n>0]} \phi \,\mathrm{d}\mu &= \int_{[\phi_n>0]\cap[\phi_n\circ f<0]} \phi \,\mathrm{d}\mu + \int_{[\phi_n>0]\cap[\phi_n\circ f\geqslant 0]} \phi \,\mathrm{d}\mu \\
&\geqslant \int_{[\phi_n>0]\cap[\phi_n\circ f<0]} \phi \,\mathrm{d}\mu + \int_{[\phi_n>0]\cap[\phi_n\circ f\geqslant 0]} \phi_n - \phi_n\circ f \,\mathrm{d}\mu \\
&= \int_{[\phi_n>0]\cap[\phi_n\circ f<0]} \phi_n \,\mathrm{d}\mu + \int_{[\phi_n>0]\cap[\phi_n\circ f\geqslant 0]} \phi_n - \phi_n\circ f \,\mathrm{d}\mu \\
&= \int_{[\phi_n>0]} \phi_n \,\mathrm{d}\mu - \int_{[\phi_n>0]\cap[\phi_n\circ f\geqslant 0]} \phi_n\circ f \,\mathrm{d}\mu \\
&\geqslant \int_{[\phi_n>0]} \phi_n \,\mathrm{d}\mu - \int_{[\phi_n\circ f\geqslant 0]} \phi_n\circ f \,\mathrm{d}\mu \\
&= \int_{[\phi_n>0]} \phi_n \,\mathrm{d}\mu - \int_{[\phi_n\geqslant 0]} \phi_n \,\mathrm{d}(f_\star\mu) = 0.
\end{aligned}$$

□

Corollary 8.11. *If $(X,\mathscr{B},\mu)$, f, α and ϕ are as in Theorem 8.10 and $A\in\mathscr{B}$ satisfies $A\subset E^\alpha(\phi)$ and $f^{-1}(A)=A$, then it holds*

$$\int_A \phi \,\mathrm{d}\mu \geqslant \alpha\mu(A).$$

Exercise 26. *Prove Corollary 8.11.*

Now we are ready for

Proof of Theorem 8.7. Let us start considering the functions $\overline{\phi},\underline{\phi}\colon X\to[-\infty,+\infty]$ given by

$$\overline{\phi}(x) := \limsup_{n\to+\infty} \mathcal{S}_f^n\phi(x) \quad\text{and}\quad \underline{\phi}(x) := \liminf_{n\to+\infty} \mathcal{S}_f^n\phi(x),$$

for every $x\in X$.

Exercise 27. *Using Lemma 8.9, show that the functions $\overline{\phi}$ and $\underline{\phi}$ are (essentially) f-invariant.*

Our purpose now consists in showing that the functions $\overline{\phi}$ and $\underline{\phi}$ coincide μ-a.e. In order to prove that, given real numbers $\alpha<\beta$, consider the set

$$E_\alpha^\beta := \left\{x\in X : \underline{\phi}(x)<\alpha<\beta<\overline{\phi}(x)\right\},$$

and observe that

$$[\underline{\phi}\neq\overline{\phi}] = [\underline{\phi}<\overline{\phi}] = \bigcup_{\alpha,\beta\in\mathbb{Q},\ \alpha<\beta} E_\alpha^\beta$$

So, to prove that $\underline{\phi}$ and $\overline{\phi}$ coincide μ-a.e. it is enough to show that $\mu\big(E_\alpha^\beta\big)=0$, for every pair of real numbers $\alpha<\beta$.

Since $\underline{\phi}$ and $\overline{\phi}$ are f-invariant, it is easy to show that $f^{-1}\Big(E_\alpha^\beta\Big) = E_\alpha^\beta$ up to sets of μ-measure zero. Then, observing that $E_\alpha^\beta\subset E^\beta(\phi)$, where the last set is defined as in Theorem 8.10, applying Corollary 8.11 we can conclude

$$\int_{E_\alpha^\beta} \phi \,\mathrm{d}\mu \geqslant \beta\mu\Big(E_\alpha^\beta\Big).$$

On the other hand, notice that $E_\alpha^\beta \subset E^{-\alpha}(-\phi)$. So, we can again apply Corollary 8.11 to the function $-\phi$ to get

$$\int_{E_\alpha^\beta} -\phi \, \mathrm{d}\mu \geqslant -\alpha\mu\left(E_\alpha^\beta\right).$$

Putting together these estimates we conclude

$$\beta\mu\left(E_\alpha^\beta\right) \leqslant \alpha\mu\left(E_\alpha^\beta\right),$$

and since $\alpha < \beta$, this can only happend when $\mu\left(E_\alpha^\beta\right) = 0$.

The fact that $\tilde{\phi} := \underline{\phi} = \overline{\phi}$ belongs to $L^1(X, \mathscr{B}, \mu)$ and

$$\int_X \phi \, \mathrm{d}\mu = \int_X \tilde{\phi} \, \mathrm{d}\mu,$$

will be proved in Corollary 8.13 □

There are some important consequences of Birkhoff ergodic theorem:

Corollary 8.12. *If $(X, \mathscr{B}, \mu)$ and f are as in Theorem 8.7 and $\phi \in L^p(X, \mathscr{B}, \mu)$ with $1 \leqslant p < \infty$, then the limit function $\tilde{\phi}$ given by Theorem 8.7 belongs to $L^p(X, \mathscr{B}, \mu)$ and it holds*

$$\left\|\tilde{\phi}\right\|_p \leqslant \|\phi\|_p .$$

Proof. Notice that

$$|\tilde{\phi}(x)| \leqslant \lim_{n\to+\infty} \frac{1}{n} \sum_{j=0}^{n-1} \left|\phi \circ f^j(x)\right|, \quad \text{for } \mu\text{-a.e. } x \in X,$$

and hence,

$$|\tilde{\phi}(x)|^p \leqslant \lim_{n\to+\infty} \left(\frac{1}{n} \sum_{j=0}^{n-1} \left|\phi \circ f^j(x)\right|\right)^p.$$

Then, as a consequence of Fatou lemma and Minkowski inequality we get

$$\begin{aligned}\left\|\tilde{\phi}\right\|_p = \left[\int_X |\tilde{\phi}(x)|^p \, \mathrm{d}\mu\right]^{\frac{1}{p}} &\leqslant \liminf_n \left[\int_X \left(\frac{1}{n} \sum_{j=0}^{n-1} \left|\phi\left(f^j(x)\right)\right|\right)^p \mathrm{d}\mu\right]^{\frac{1}{p}} \\ &\leqslant \liminf_n \frac{1}{n} \sum_{j=0}^{n-1} \left(\int_X \left|\phi\left(f^j(x)\right)\right|^p \mathrm{d}\mu\right)^{\frac{1}{p}} = \|\phi\|_p .\end{aligned}$$

□

Corollary 8.13 (L^p-convergence of ergodic averages)**.** *If $(X, \mathscr{B}, \mu)$, f, ϕ and $\tilde{\phi}$ are as in Corollary 8.12, then*

$$\left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_p \to 0,$$

as $n \to +\infty$, and where $\mathcal{S}_f^n \phi = \frac{1}{n} \sum_{j=0}^{n-1} \phi \circ f^j$. Therefore, it holds

$$\int_X \phi \, \mathrm{d}\mu = \int_X \tilde{\phi} \, \mathrm{d}\mu.$$

Proof. By Corollary 8.13 we know that $\tilde{\phi} \in L^p(X, \mathscr{B}, \mu)$. Then to prove the converges in L^p, let us start assuming $\phi \in L^\infty(x, \mathscr{B}, \mu)$. So, $\left\|\mathcal{S}_f^n \phi\right\|_\infty \leqslant \|\phi\|_\infty$, for every $n \geqslant 1$, and hence, $\tilde{\phi} \in L^\infty(X, \mathscr{B}, \mu)$, as well. So,

$$\left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_p \leqslant \left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_\infty \leqslant \|\phi\|_\infty + \|\tilde{\phi}\|_\infty,$$

and $\mathcal{S}_f^n \phi \to \tilde{\phi}$, μ-a.e. as $n \to \infty$. So, by dominated convergence theorem, it holds $\left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_p \to 0$, as $n \to \infty$.

Now, to prove the convergence in L^p without the additional assumption that ϕ is bounded, let us consider a sequence of truncations of ϕ, i.e. for each $k \in \mathbb{N}$ let us consider the function $\phi^{(k)}(x) := \phi(x)\mathbb{1}_{\phi^{-1}[-k,k]}(x)$. Observe that $\phi^{(k)} \in L^\infty(X, \mathscr{B}, \mu)$ for every k and $\phi^{(k)} \to \phi$ as $k \to \infty$, μ-almost everywhere and in L^p.

So, given any $\varepsilon > 0$ there exists $k \in \mathbb{N}$ such that $\left\|\phi^{(k)} - \phi\right\|_p < \varepsilon/3$, and hence, $\left\|\mathcal{S}_f^n(\phi^{(k)} - \phi)\right\|_p < \varepsilon/3$, for every $n \geqslant 1$, and by Corollary 8.13, it holds

$$\left\|\tilde{\phi}^{(k)} - \tilde{\phi}\right\|_p \leqslant \left\|\phi^{(k)} - \phi\right\|_p < \frac{\varepsilon}{3}.$$

Then, by the previous argument, we have $\left\|\mathcal{S}_f^n \phi^{(k)} - \tilde{\phi}^{(k)}\right\|_p < \varepsilon/3$, whenever n is sufficiently large, and so we get

$$\begin{aligned}\left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_p &\leqslant \left\|\mathcal{S}_f^n \phi - \mathcal{S}_f^n \phi^{(k)}\right\|_p + \left\|\mathcal{S}_f^n \phi^{(k)} - \tilde{\phi}^{(k)}\right\|_p + \left\|\tilde{\phi}^{(k)} - \tilde{\phi}\right\|_p \\ &\leqslant \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,\end{aligned}$$

for n large enough.

Finally, in order to prove that $\int_x \phi \, \mathrm{d}\mu = \int_X \tilde{\phi} \, \mathrm{d}\mu$, it is enough to notice that if $\phi \in L^p(X, \mathscr{B}, \mu)$, then $\phi \in L^1(X, \mathscr{B}, \mu)$ and it holds

$$\begin{aligned}\left|\int_X \phi \, \mathrm{d}\mu - \int_X \tilde{\phi} \, \mathrm{d}\mu\right| &= \left|\int_X \mathcal{S}_f^n \phi \, \mathrm{d}\mu - \int_X \phi \, \mathrm{d}\mu\right| \\ &\leqslant \int_X \left|\mathcal{S}_f^n \phi - \tilde{\phi}\right| \, \mathrm{d}\mu = \left\|\mathcal{S}_f^n \phi - \tilde{\phi}\right\|_1 \to 0,\end{aligned}$$

as $n \to +\infty$. □

Exercise 28. *Show that Corollary 8.13 is false for $p = \infty$.*

Corollary 8.14 (Birkhoff theorem for continuous functions)**.** *Let X be compact metric space and $f\colon X \circlearrowleft$ be a continuous map. Then there exists a Borel subset $X' \subset X$ such that $\mu(X') = 1$, for every $\mu \in \mathfrak{M}(f)$ and the following limit exists for every $x \in X'$ and every $\phi \in C^0(X, \mathbb{R})$:*

$$\tilde{\phi}(x) := \lim_{n \to +\infty} \frac{1}{n} \sum_{j=0}^{n-1} \phi\big(f^j(x)\big).$$

Proof. Since X is compact, the Banach space $C^0(X, \mathbb{R})$ is separable, i.e. there exists a sequence $(\phi_k)_{k \geqslant 1}$ which is dense in $C^0(X, \mathbb{R})$. For each $k \in \mathbb{N}$ we define

$$X_k := \left\{ x \in X : \lim_{n \to +\infty} \frac{1}{n} \sum_{j=0}^{n-1} \phi_k\big(f^j(x)\big) \text{ exists} \right\}.$$

The set X_k is clearly a Borel set and, by Birkhoff theorem (Theorem 8.7) we know that $\mu(X_k) = 1$, for every $k \in \mathbb{N}$ and all $\mu \in \mathfrak{M}(f)$. So, if we define

$$X' := \bigcap_{k\in\mathbb{N}} X_k,$$

it clearly holds $\mu(X') = 1$, for every $\mu \in \mathfrak{M}(X)$.

Then, let us consider an arbitrary point $x \in X'$ and a positive real number $\varepsilon > 0$. Given any $\phi \in C^0(X,\mathbb{R})$, by denseness of the sequence (ϕ_k) we know there exists $k \in \mathbb{N}$ such that $\|\phi - \phi_k\|_0 < \varepsilon/2$. So we have

$$\begin{aligned}
0 &\leqslant \limsup_n \frac{1}{n}\sum_{j=0}^{n-1} \phi\big(f^j(x)\big) - \liminf_n \frac{1}{n}\sum_{j=0}^{n-1} \phi\big(f^j(x)\big) \\
&\leqslant \left|\limsup_n \frac{1}{n}\sum_{j=0}^{n-1} \phi\big(f^j(x)\big) - \limsup_n \frac{1}{n}\sum_{j=0}^{n-1} \phi_k\big(f^j(x)\big)\right| \\
&\qquad + \left|\liminf_n \frac{1}{n}\sum_{j=0}^{n-1} \phi_k\big(f^j(x)\big) - \liminf_n \frac{1}{n}\sum_{j=0}^{n-1} \phi\big(f^j(x)\big)\right| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
\end{aligned}$$

Therefore, the $lim_n \frac{1}{n}\sum_{j=0}^{n-1} \phi(f^j(x))$ exists and the result is proved. □

9. Ergodicity

Let $(X,\mathscr{B},\mu)$ be a probability space and $f\colon X \circlearrowleft$ be a measure preserving map. We say that (f,μ) is **ergodic** when given any $A \in \mathscr{B}$ such that $f^{-1}(A) = A$, either $\mu(A) = 0$ or $\mu(A) = 1$.

Ergodic systems satisfy the following properties:

Theorem 9.1. *Let $(X,\mathscr{B},\mu)$ be a probability space and $f\colon X \circlearrowleft$ be a measure preserving map. Then the following statements are equivalent:*

(1) (f,μ) is ergodic;

(2) given any $B \in \mathscr{B}$ such that $\mu\big(B \triangle f^{-1}(B)\big) = 0$, then $\mu(B) \in \{0,1\}$;

(3) if $\phi \in X \to \mathbb{R}$ is a measurable function such that

$$\phi\big(f(x)\big) = \phi(x), \quad \text{for } \mu\text{-a.e. } x \in X,$$

then there exists $c \in \mathbb{R}$ such that $\phi(x) = c$, for μ-a.e. $x \in X$;

(4) for every $A, B \in \mathscr{B}$ it holds

$$\frac{1}{n}\sum_{j=0}^{n-1} \mu\big(f^{-j}(A) \cap B\big) \to \mu(A)\mu(B),$$

as $n \to +\infty$;

(5) given $p,q \in (1,\infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$, and $\phi \in L^p(X,\mathscr{B},\mu)$ and $\psi \in L^q(X,\mathscr{B},\mu)$, it holds

$$\frac{1}{n}\sum_{j=0}^{n-1} \int_X (\phi \circ f^j)\psi \,\mathrm{d}\mu \to \int_X \phi \,\mathrm{d}\mu \int_X \psi \,\mathrm{d}\mu,$$

as $n \to +\infty$.

Exercise 29. *Let $(X,\mathscr{B},\mu)$ be a probability space and $f\colon X \circlearrowleft$ be measure preserving map. Let $B\in\mathscr{B}$ be a measurable set such that $\mu\big(B\triangle f^{-1}(B)\big)=0$. Show that there exists $A\in\mathscr{B}$ satisfying $f^{-1}(A)=A$ and $\mu(A\triangle B)=0$.*

Proof of Theorem 9.1. Clearly *(2)* implies *(1)*. On the other hand, by Exercise 29 we know that *(1)* implies *(2)*.

If (f,μ) is not ergodic, then there exists $A\in\mathscr{B}$ with $f^{-1}(A)=A$ and $0<\mu(A)<1$. Then, $\mathbb{1}_A\circ f=\mathbb{1}_{f^{-1}(A)}=\mathbb{1}_A$ and $\mathbb{1}_A$ is not constant μ-a.e. So, *(3)* implies *(1)*. On the other hand, if $\phi\big(f(x)\big)=\phi(x)$ for μ-a.e. point $x\in X$ and ϕ is not constant μ-almost everywhere, then there exists $r\in\mathbb{R}$ such that $B:=\phi^{-1}[-\infty,r]$ satisfies $0<\mu(B)<1$, with $\mu\big(B\triangle f^{-1}(B)\big)=0$. Thus, *(2)* implies *(3)*.

Then, if *(4)* holds and $A\in\mathscr{B}$ is f-invariant, then we have

$$\mu(A)=\frac{1}{n}\sum_{j=0}^{n-1}\mu\big(A\cap f^{-1}(A)\big)\to\mu(A)^2,\quad\text{as } n\to+\infty.$$

So $\mu(A)\in\{0,1\}$ and *(1)* holds.

On the other hand, assuming *(5)* we can easily prove *(4)* taking $\phi:=\mathbb{1}_A$ and $\psi:=\mathbb{1}_B$.

Finally, let us prove that *(1)* implies *(5)*. To do that, let us star assuming $\phi\in L^\infty(X,\mathscr{B},\mu)$ and $\psi\in L^q(X,\mathscr{B},\mu)$. Then, by Birkhoff ergodic theorem (Theorem 8.7), and since *(1)* implies *(3)*, we know that

$$\frac{1}{n}\sum_{i=0}^{n-1}\phi\big(f^i(x)\big)\to\int_X\phi\,\mathrm{d}\mu,\quad\text{as } n\to+\infty,$$

and for μ-a.e. $x\in X$, and we have

$$\left|\frac{1}{n}\sum_{i=0}^{n-1}\phi\big(f^i(x)\big)\psi(x)\right|\leqslant\|\phi\|_\infty\,|\psi(x)|\,,\quad\forall n\in\mathbb{N},$$

where $|\psi|$ is an integrable function. So, by dominated converge theorem we conclude that

$$\lim_{n\to+\infty}\frac{1}{n}\sum_{i=0}^{n-1}\int_x(\phi\circ f^i)\psi\,\mathrm{d}\mu=\left(\int_x\phi\,\mathrm{d}\mu\right)\left(\int_X\psi\,\mathrm{d}\mu\right),$$

we wanted to prove. The general case, where $\phi\in L^p(X,\mathscr{B},\mu)$ can be proved taking the sequence of truncations of ϕ given by $\phi_k:=\mathbb{1}_{[|\phi|\leqslant k]}\phi\in L^\infty(X,\mathscr{B},\mu)$, for every $k\in\mathbb{N}$. Then we have $\phi_k\to\phi$ in L^p and μ-almost everywhere. Then, given any $\varepsilon>0$ there is $k\in\mathbb{N}$ such that $\|\phi-\phi_k\|_p<\varepsilon$ and, by the previous argument, there is $N\in\mathbb{N}$ such that

$$\left|\frac{1}{n}\sum_{i=0}^{n-1}\int_X(\phi\circ f^i)\psi\,\mathrm{d}\mu-\int_X\psi\,\mathrm{d}\mu\int_X\psi\,\mathrm{d}\mu\right|<\varepsilon,\quad\forall n\geqslant N.$$

Hence we have

$$\begin{aligned}
&\left|\frac{1}{n}\sum_{i=0}^{n-1}\int_X(\phi\circ f^i)\psi\,\mathrm{d}\mu-\int_X\phi\,\mathrm{d}\mu\int_X\psi\,\mathrm{d}\mu\right|\\
&\leqslant\left|\frac{1}{n}\sum_{i=0}^{n-1}\int_X\big((\phi-\phi_k)\circ f^i\big)\psi\,\mathrm{d}\mu\right|+\left|\frac{1}{n}\sum_{i=0}^{n-1}\int_X(\phi_k\circ f^i)\psi\,\mathrm{d}\mu-\int_X\phi_k\,\mathrm{d}\mu\int_X\psi\,\mathrm{d}\mu\right|\\
&\qquad+\left|\int_X(\phi_k-\phi)\,\mathrm{d}\mu\int_X\psi\,\mathrm{d}\mu\right|\leqslant\|\phi-\phi_k\|_p\,\|\psi\|_q+\varepsilon+\|\phi_k-\phi\|_1\,\|\psi\|_1\\
&\leqslant\varepsilon\,\|\psi\|_q+\varepsilon+\|\phi_k-\phi\|_p\,\|\psi\|_q\leqslant\varepsilon\big(2\,\|\psi\|_q+1\big),
\end{aligned}$$

for every $n \geqslant N$. □

We shall need a little improvement of condition *(3)* of Theorem 9.1:

Exercise 30. *Let $(X, \mathscr{B}, \mu)$ be a probability space and $f\colon X \circlearrowleft$ be a measure preserving map, and let $\mathcal{A} \subset 2^X$ be an algebra such that $\mathscr{B}$ is the σ-algebra generated by $\mathcal{A}$ and suppose that*

$$\frac{1}{n}\sum_{i=0}^{n-1} \mu\big(f^{-i}(A) \cap B\big) \to \mu(A)\mu(B), \quad \text{as } n \to +\infty,$$

for every $A, B \in \mathcal{A}$. Then, show that (f, μ) is ergodic.

9.1. **Examples of ergodic systems.**

9.1.1. *Circle rotations.* Given any $\alpha \in \mathbb{R}$, we define the rotation $T_\alpha\colon \mathbb{R}/\mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ by

$$T_\alpha(x + \mathbb{Z}) := x + \alpha + \mathbb{Z}, \quad \forall x \in \mathbb{R}.$$

Since $\mathbb{R}/\mathbb{Z}$ is a compact group, its Haar measure λ is T_α-invariant for every $\alpha \in \mathbb{R}$.

Exercise 31. *Let $\pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{Z}$ be a the natural quotient map, i.e. $\pi(x) := x + \mathbb{Z}$, for every $x \in \mathbb{R}$. Show that $\lambda = \pi_\star\Big(\mathrm{Leb}\big|_{[0,1]}\Big)$.*

Exercise 32. *Given $\alpha \in \mathbb{Q}$, show that (T_α, λ) is not ergodic.*

Theorem 9.2. *The system (T_α, λ) is ergodic for every $\alpha \in \mathbb{R}\backslash\mathbb{Q}$.*

9.1.2. *Expanding maps on the interval.* Given any $k \in \mathbb{N}$ with $k \geqslant 2$, consider the map $f\colon [0,1] \circlearrowleft$ given by

$$f(x) := kx - \lfloor kx \rfloor, \quad \forall x \in [0,1].$$

We have shown in §5.2 that the measure $\lambda := \mathrm{Leb}\big|_{[0,1]}$ is f-invariant. Moreover, we have

Theorem 9.3. *The system (f, λ) is ergodic.*

9.1.3. *Bernoulli shifts.*

10. Unique ergodicity

11. Ergodic decomposition

11.1. **Ergodic measures as extreme points.** Let $(X, \mathscr{B})$ be a measurable space and $f\colon X \circlearrowleft$ be a measurable map. We first observe that ergodic measures are "minimal elements" with respect to the preorder given by absolute continuity among invariant measures:

Proposition 11.1. *If μ and ν are two f-invariant probability measures such that μ is ergodic and $\nu \ll \mu$ (i.e. $\nu(A) = 0$, for every $A \in \mathscr{B}$ such that $\mu(A) = 0$), then $\nu = \mu$.*

Proof. Given an arbitrary set $A \in \mathscr{B}$, by Birkhoff ergodic theorem and since μ is ergodic, we get that there exists a measure $R_A \in \mathscr{B}$ with $\mu(R_A) = 1$ such that

$$\frac{1}{n}\sum_{i=0}^{n-1} \mathbb{1}_A \circ f^j(x) \to \mu(A), \quad \text{as } n \to +\infty,\ \forall x \in R_A.$$

Since $\nu \ll \mu$ and $\mu(X\backslash R_A) = 0$, we get $\nu(X\backslash R_A) = 0$. So the above Birkhoff averages converge to $\mu(A)$ for ν-a.e. $x \in \mathbb{R}$ and consequently, $\mu(A) = \nu(A)$. Since A was arbitrary, we get $\mu = \nu$. □

We get the following

Corollary 11.2. *Let $(X,\mathscr{B})$ be a measurable space and $f\colon X \circlearrowleft$ be a measurable map. If μ denotes an f-invariant probability measure, then μ is not ergodic if and only there exist two f-invariant measures ν_1 and ν_2 with $\nu_1 \neq \nu_2$ and $t \in (0,1)$ such that*

$$\mu = t\nu_1 + (1-t)\nu_2.$$

Proof. Let us start assumung μ is not ergodic. Then there is $A \in \mathscr{B}$ with $0 < \mu(A) < 1$. So, let us define $\nu_!(E) := \dfrac{1}{\mu(A)}\mu(A \cap E)$ and $\nu_2 := \dfrac{1}{\mu(X\backslash A)}\mu(E\backslash A)$, for every $E \in \mathscr{B}$, and it clearly holds

$$\mu = \mu(A)\nu_1 + \mu(X\backslash A)\nu_2 = \mu(A)\nu_1 + \big(1-\mu(A)\big)\nu_2.$$

Reciprocally, let us suppose that μ is ergodic and there are f-invariant measures ν_1 and ν_2 and $t \in (0,1)$ such that $\mu = t\nu_1 + (1-t)\nu_2$. Then it clearly holds $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$. By Proposition 11.1, we know that $\mu = \nu_1 = \nu_2$. □

11.2. **Partitions of measure spaces.** Let $(X,\mathscr{B},\mu)$ be an arbitrary probability space. A **partition** of X is a family $\xi \subset \mathscr{B}$ such that $C \cap D = \varnothing$, for every $C, D \in \xi$ with $C \neq D$ and

$$\mu\Big(\bigsqcup_{C\in\xi} C\Big) = 1.$$

The elements of ξ are called **atoms** of ξ. Given a point $x \in \bigsqcup_{C\in\xi} C$, we define $\xi(x) \in \xi$ as the only atom of ξ that contains the point x. So, we can define the natural *projection map* $\pi^\xi\colon X \to \xi$ by $\pi^\xi(x) := \xi(x)$ and notice that π^ξ is well defined μ-almost everywhere.

We can use the natural projection map π^ξ to endow the set ξ with a measure space structure. Indeed, we shall always assume the partition ξ as the supporting set of the measure space $(\xi,\mathscr{B}_\xi,\hat{\mu}^\xi)$ where $\mathscr{B}_\xi := \pi^\xi_\star\mathscr{B}$ and $\hat{\mu}^\xi := \pi^\xi_\star\mu$ denote the push-forward σ-algebra of $\mathscr{B}$ and the measure μ by π^ξ, respectively.

We can define a partial order relation among partitions of a measure space. Given two partitions ξ,η of the measure space $(X,\mathscr{B},\mu)$, we say that ξ is **finer** than η, or that η is **coarser** that ξ, and we write $\eta \leqslant \xi$, when there is a measurable subset $X' \subset X$ with $\mu(X') = 1$ and such that

$$\xi(x) \cap X' \subset \eta(x) \cap X', \quad \forall x \in \Big(\bigsqcup_{C\in\xi} C\Big) \cap \Big(\bigsqcup_{D\in\eta} D\Big).$$

We say that ξ and η coincide or are **equal** when $\eta \leqslant \xi$ and $\xi \leqslant \eta$.

If we write $\mathcal{P}_X$ to denote the set of all partitions of the measure space $(X,\mathscr{B},\mu)$, we know that $(\mathcal{P}_X,\leqslant)$ is a partially order set.

Exercise 33. *Show that the notions of* infimum *and* supremum *are well defined for countable subsets on the partially order set $(\mathcal{P}_X,\leqslant)$. More precisely, show that given any countable family of partitions $\Xi \subset \mathcal{P}_X$, there exists two partitions $\inf\Xi, \sup\Xi \in \mathcal{P}_X$ such that*

$$\eta \leqslant \inf\Xi \leqslant \xi \leqslant \sup\Xi \leqslant \eta', \quad \forall \xi \in \Xi,$$

and every $\eta,\eta' \in \mathcal{P}_X$ such that $\eta \leqslant \xi' \leqslant \eta'$, for every $\xi' \in \mathcal{P}_X$.

We can use the notions of infimum and supremum on $\mathcal{P}_X$ to define the operators $\wedge$ and $\vee$: given any countable (or finite) family $\Xi \subset \mathcal{P}_X$, we define

$$\bigwedge_{\xi\in\Xi} \xi := \sup\{\eta \in \mathcal{P}_X : \eta \leqslant \xi,\ \forall \xi \in \Xi\},$$

and

$$\bigvee_{\xi\in\Xi}\xi := \inf\{\eta\in\mathcal{P}_X : \xi\leqslant\eta,\ \forall\xi\in\Xi\}. \tag{12}$$

11.3. **Disintegration of measures.** Given a probability space $(X,\mathscr{B},\mu)$ and a partion $\xi\in\mathcal{P}_X$, a **disintegration** of μ with respect to ξ is family $\{\mu_C : C\in\xi\}$ satisfying the following properties:

(i) μ_C is a probability measure on the measurable space $(X,\mathscr{B})$, for every $C\in\xi$;
(ii) $\mu_C(C)=1$, for $\hat{\mu}^\xi$-almost every atom $C\in\xi$;
(iii) for every $E\in\mathscr{B}$, the map $\xi\ni C\mapsto\mu_C(E)\in[0,1]$ is $(\mathscr{B}^\xi,\mathscr{B}_{\mathbb{R}})$-measurable;
(iv) it holds

$$\mu(E)=\int_\xi \mu_C(E)\,\mathrm{d}\hat{\mu}^\xi(C),\quad \forall E\in\mathscr{B}.$$

Let us observe that one can pull back the partion and push forward the disintegrations:

Exercise 34. *Let $(X,\mathscr{B},\mu)$ be a probability space, $\xi\in\mathcal{P}_X$ be a partion and $f\colon X\circlearrowleft$ be a measure preserving map. Then, show that $f^\star\xi:=\{f^{-1}(C) : C\in\xi\}$ is a partition of $(X,\mathscr{B},\mu)$. On the other hand, if $\{\mu_D : D\in f^\star\xi\}$ is a disintegration of μ with respect to the partition $f^\star\xi$, show that $\{\mu_C := f_\star\mu_{f^{-1}(C)} : C\in\xi\}$ is a disintegration of μ with respect to ξ.*

First we shall prove that every partition admits at most one disintegration:

Theorem 11.3 (Uniqueness of disintegrations)**.** *If $(X,\mathscr{B},\mu)$ is a Borel space (see §3), $\xi\in\mathcal{P}_X$ is a partition, and $\{\mu_C : C\in\xi\}$ and $\{\nu_C : C\in\xi\}$ are two disintegrations of μ with respect to ξ, then it holds*

$$\mu_C=\nu_C,\quad \text{for } \hat{\mu}^\xi\text{-a.e. } C\in\xi.$$

Proof. Since $(X,\mathscr{B},\mu)$ is a Borel space, there exists a countable algebra $\mathscr{A}\subset\mathscr{B}$ that generates $\mathscr{B}$. For each $A\in\mathscr{A}$, let us consider the sets

$$\mathscr{M}_A:=\{C\in\xi : \mu_C(A)<\nu_C(A)\},\quad \text{and } \mathscr{N}_A:=\{C\in\xi : \mu_C(A)>\nu_C(A)\}.$$

Then, since $\{\mu_C : C\in\xi\}$ is a disintegration of μ with respect to ξ, we have

$$\begin{aligned}\mu\big(A\cap\pi^{\xi^{-1}}(\mathscr{M}_A)\big)&=\int_\xi \mu_C\big(A\cap\pi^{\xi^{-1}}(\mathscr{M}_A)\big)\,\mathrm{d}\hat{\mu}^\xi(C)\\&=\int_{\mathscr{M}_A}\mu_C(A)\,\mathrm{d}\hat{\mu}^\xi(C).\end{aligned}$$

Analogously, one can show that

$$\mu\big(A\cap\pi^{\xi^{-1}}(\mathscr{M}_A)\big)=\int_{\mathscr{M}_A}\nu_C(A)\,\mathrm{d}\hat{\mu}^\xi(C).$$

Since $\mu_C(A)<\nu_C(A)$, for all $C\in\mathscr{M}_A$, we get $\hat{\mu}^\xi(\mathscr{M}_A)=0$. Analogously, we can show that $\hat{\mu}^\xi(\mathscr{N}_A)=0$ and hence, $\mu_C=\nu_C$, for $\hat{\mu}^\xi$-almost every $C\in\xi$. □

Theorem 11.3 can be used to show that certain partitions do not admit any disintegration of the measure:

Example 11.4. Let α be an arbitrary real irrational number and $f\colon\mathbb{R}/\mathbb{Z}\circlearrowleft$ be the corresponding rotation $f(x+\mathbb{Z}):=x+\alpha+\mathbb{Z}$, for every $x\in\mathbb{R}$. Let λ denote the Haar measure on the circle $\mathbb{R}/\mathbb{Z}$ and consider the partition ξ of the probability space $(\mathbb{R}/\mathbb{Z},\mathscr{B}_{\mathbb{R}/\mathbb{Z}},\lambda)$ given by the orbits of f, i.e. we have

$$\xi:=\{\{f^n(x)\in\mathbb{R}/\mathbb{Z} : n\in\mathbb{Z}\} : x\in\mathbb{R}/\mathbb{Z}\}.$$

Then there is no disintegration of λ with respect to the partion ξ.

Now we can state the main result of this section:

Theorem 11.5 (Ergodic decomposition theorem)**.** *Let $(X, \mathscr{B}, \mu)$ be a Borel space and $f\colon X \circlearrowleft$ be a measure preserving map. Then there exists a unique partition $\xi_f \in \mathcal{P}_X$, called the* ***ergodic partition of*** *f, such ξ_f admits a disintegration $\{\mu_C : C \in \xi_f\}$ and the following conditions hold:*

(1) $f^\star \xi_f = \xi_f$, i.e. $f^{-1}(C) = C$, for $\hat{\mu}^{\xi_f}$-almost every atom $C \in \xi_f$;

(2) μ_C is an ergodic f-invariant measure, for $\hat{\mu}^{\xi_f}$-almost every atom $C \in \xi_f$.

The proof of Theorem 11.5 will be postpone and get as a consequence of Rokhlin decomposition theorem.

11.4. **Measurable partitions.** By Example 11.4 we know that some partitions do not admit disintegrations. In this section we shall characterize the family of partitions admitting disintegrations.

Given a probability space $(X, \mathscr{B}, \mu)$ we say that a partition $\xi \in \mathcal{P}_X$ is a **measurable partition** when there exists a sequence $(\xi_n)_{n\geqslant 1} \subset \mathcal{P}_X$ of finite partitions such that

$$\xi = \bigvee_{n\geqslant 1} \xi_n,$$

where the equality among partitions is given in §11.2 and the operator $\bigvee$ is given by (12)

Exercise 35. *Let $(X, \mathscr{B}, \mu)$ be a probability space, M be separable metric space and $f\colon X \to M$ be a measurable map. Then the partition*

$$\xi_f := \left\{ f^{-1}(y) : y \in M \right\} \in \mathcal{P}_X$$

is a measurable partion.

Then we have the main result about measurable partitions:

Theorem 11.6 (Rokhlin theorem)**.** *Let $(X, \mathscr{B}, \mu)$ be a Borel space and $\xi \in \mathcal{P}_X$ be measurable partition. Then there exits a disintegration of μ with respect to ξ.*

We postpone the proof of Theorem 11.6. Now we shall prove the Ergodic decomposition theorem assuming Theorem 11.6.

Proof of Theorem 11.5. Since $(X, \mathscr{B}, \mu)$ is a Borel space, there exists a countable algebra $\mathscr{A} \subset \mathscr{B}$ that generates the σ-algebra $\mathscr{B}$. For each $A \in \mathscr{A}$, let us consider the function $\tau_A\colon X \to [0,1]$ given by

$$\tau_A(x) := \lim_{n\to+\infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{1}_A\big(f^i(x)\big),$$

whenever the limit exists. By Birkhoff ergodic theorem we know that τ_A is well defined μ-almost everywhere.

Since $\mathscr{A}$ is countable, we get the set $X_0 := \{x \in X : \tau_A(x)$ is well defined, $\forall A \in \mathscr{A}\}$ is a measurable set with $\mu(X_0) = 1$.

Then we define the partition $\xi_f \in \mathcal{P}_X$ as the set of equivalence classes on X_0 such that $x \sim y$ if and only if $\tau_A(x) = \tau_A(y)$, for every $A \in \mathscr{A}$.

Exercise 36. *Show that ξ_f is a measurable partition.*

Then, by Rokhlin theorem (Theorem 11.6) we know that there is a disintegration of μ with respect to ξ_f $\{\mu_C : C \in \xi_f\}$.

By the very same definition of functions τ_A we know that $f^{-1}(X_0) = X_0$ and $\tau_A(x) = \tau_A \circ f(x)$, for every $x \in X_0$. So, it clearly holds $f^{-1}(C) = C$, for every

$C \in \xi_f$. Thus, by uniqueness of disintegrations (Theorem 11.3), we conclude that $f_\star \mu_C = \mu_C$, for $\hat{\mu}^{\xi_f}$-almost every atom $C \in \xi_f$.

So, it just remains to prove that the measure μ_C is ergodic for f, for every atom $C \in \xi_f$. To do that, let us fix an atom $C \in \xi_f$ and observe that to prove the ergodicity of μ_C is enough to show that $\tau_E(x) = \mu_C(E)$, for every $E \in \mathscr{B}$ and μ_C-a.e. $x \in C$. Then, let us define the family

$$\mathscr{B}_C := \{E \in \mathscr{B} : \tau_E(x) = \mu_C(E), \text{ for } \mu_C\text{-a.e. } x \in C\}.$$

By our definition it clearly holds $\mathscr{A} \subset \mathscr{B}_C$. On the other hand, if $E_1, E_2 \in \mathscr{B}_C$, and $E_1 \subset E_2$, then $E_2 \backslash E_1 \in \mathscr{B}_C$ because

$$\tau_{E_2\backslash E_1}(x) = \tau_{E_2}(x) - \tau_{E_1}(x) = \mu_C(E_2\backslash E_1),$$

for μ_C-almost every $x \in C$; and if $E_1, E_2, \ldots \in \mathscr{B}_C$ is a family of pairwise disjoint sets, the it clearly holds

$$\tau_{\bigsqcup_j E_j}(x) = \sum_j \tau_{E_j}(x) = \mu_C\left(\bigsqcup_j E_j\right).$$

Then, the ergodicity of μ_C is consequence of the following

Exercise 37. *A family $\mathcal{C} \subset 2^X$ is called a **monotone class** when given any sequence $A_1 \subset A_2 \subset \ldots$ and $B_1 \supset B_2 \supset \ldots$, with $A_n, B_n \in \mathcal{C}$, for every $n \in \mathbb{N}$, it holds $\bigcup_n A_n, \bigcap_n B_n \in \mathcal{C}$.*

Then, if $\mathcal{C}$ is a monotone class and contains an algebra $\mathscr{A}$, then $\mathcal{C}$ contains the σ-algebra $\sigma(\mathscr{A})$ generated by $\mathscr{A}$.

□

11.5. **Conditional expectations.** In order to prove Theore 11.6 we first need to consider the notion of **conditional expectations.** Let $(X, \mathscr{B}, \mu)$ denote an arbitrary probability space and let $\mathscr{A} \subset \mathscr{B}$ be a sub-σ-algebra of $\mathscr{B}$. A real function $\phi\colon X \to \mathbb{R}$ is said to be $\mathscr{A}$-measurable when $\phi^{-1}(E) \in \mathscr{A}$ for every measurable subset $E \subset \mathbb{R}$.

Theorem 11.7 (Conditional expectation)**.** *There exists a continuous linear operator $\mathbb{E}(\cdot|\mathscr{A})\colon L^1(X, \mathscr{B}, \mu) \to L^1(X, \mathscr{A}, \mu)$ such that*

$$\int_A \phi \, \mathrm{d}\mu = \int_A \mathbb{E}(\phi|\mathscr{A}) \, \mathrm{d}\mu, \quad \forall A \in \mathscr{A}.$$

Proof of Theorem 11.7. Given an arbitrary function $\phi\colon X \to \mathbb{R}$, we can write $\phi = \phi^+ - \phi^-$, where $\phi^+ := \max(\phi, 0)$ and $\phi^- := \max(-\phi, 0)$. So, in order to define the linear operator $\mathbb{E}(\cdot|\mathscr{A})$ it is enough to define it on non-negative functions.

So, let us suppose $\phi \in L^1(X, \mathscr{B}, \mu)$ and $\phi \geqslant 0$. Consider the measure ν_ϕ on the measurable space $(X, \mathscr{A})$ given by

$$\nu_\phi(A) := \int_A \phi \, \mathrm{d}\mu, \quad \forall A \in \mathscr{A}.$$

Since $\nu_\phi \ll \mu\big|_{\mathscr{A}}$, by Radon-Nikodym theorem we know that there is $\mathbb{E}(\phi|\mathscr{A}) \in L^1(X, \mathscr{A}, \mu)$ such that

$$\nu_\phi(A) = \int_A \phi \, \mathrm{d}\mu = \int_A \mathbb{E}(\phi|\mathscr{A}) \, \mathrm{d}\mu, \quad \forall A \in \mathscr{A}.$$

Exercise 38. *Show that $\mathbb{E}(\cdot|\mathscr{A})$ can be uniquely extended as a linear operator to the whole space $L^1(X, \mathscr{B}, \mu)$ and this extension is continuous.*

□

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